SURROGATE-ASSISTED DOUBLE MACHINE LEARNING: CAUSAL INFERENCE WITH BINARY AND ORDINAL OUTCOMES

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Brandon M. Greenwell
Department of Operations, Business Analytics, and Information Systems
University of Cincinnati Carl H. Lindner College of Business
Cincinnati, Ohio, 45221
greenwbm@ucmail.uc.edu

Other Authors
University of XYZ
other.authors@email.com

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ABSTRACT

Double Machine Learning (DML) pairs Neyman-orthogonal scores with cross-fitting to estimate causal effects under flexible adjustment. For categorical outcomes, existing DML methods can target probability- or distribution-scale contrasts, but the partially linear latent-utility coefficient requires different machinery. A binary logistic estimator supplies that coefficient under a logit link; we are aware of no ordinal counterpart. We introduce SUDO (Surrogate-Assisted DML), which treats the latent outcome behind a binary or ordinal response as missing data. Each observation is completed to a continuous *surrogate* drawn from the assumed error distribution and truncated to the interval implied by its observed category, after which ordinary partialling-out applies. The completions form multiple imputations. In the parametric designs studied, Rubin pooling attained near-nominal coverage only with *proper* draws that redraw the imputation model's parameters, including ordinal thresholds; naive and improper draws under-covered. The method accommodates multiple links, and its surrogate residuals provide a link diagnostic. We also derive how imputation-model error reaches the target: it enters at first order through a truncation factor that rises from about 0.3 to 1 as the signal strengthens. For a deterministic logistic-series learner, we prove a projected expansion and full-pipeline bootstrap result. In the machine-learning designs studied, normal intervals based on a full-pipeline bootstrap attained nominal coverage for generalized additive model (GAM), multivariate adaptive regression spline (MARS), and neural-backfit imputation learners; extending the proof to these adaptive algorithms remains open. We illustrate the method on wine-quality data by estimating the latent-scale effect of volatile acidity on an ordinal quality score.

1 Introduction

Many outcomes that scientists want causal effects for are categorical. A patient relapses or not; a wine is rated on a five- or ten-point scale; a survey respondent is dissatisfied, neutral, or satisfied. For a *continuous* outcome, Double Machine Learning (DML, Chernozhukov et al. 2018) provides a now-standard route to a causal effect under flexible, high-dimensional adjustment. It predicts the outcome and treatment from the covariates with machine-learning (ML) models, removes both predictions, and reads the effect from a residual-on-residual regression. This Frisch-Waugh-Lovell (FWL) orthogonalization allows flexible nuisance predictions because, under standard convergence-rate conditions, the Neyman-orthogonal score is insensitive to their first-order estimation error (Chernozhukov et al. 2018).

That partially linear machinery is built for an observed continuous outcome. DML can be applied to categorical outcomes through other scores, but those procedures target probability- or distribution-scale contrasts. A binary or ordinal outcome obeys the additive structure used here only on a *latent* scale. An unobserved continuous utility follows $U = \theta_0 D + g_0(X) + \varepsilon$, while the data reveal only the interval containing U. Applying continuous partialling-out to the category labels is computationally possible, but it changes the estimand: the result is a probability-scale contrast rather than θ_0.

For binary outcomes, the logistic partially linear model of M. Liu, Zhang, and Zhou (2021) provides one remedy and has an official implementation in the DoubleML software. Its orthogonal score builds in the logit link, confining the method to binary responses under a logistic distribution. We are aware of no corresponding DML estimator for ordinal latent-utility effects. Ordinal latent-scale outcomes are therefore the clearest gap addressed here. Under a logistic link, SUDO targets the same latent-scale coefficient as M. Liu, Zhang, and Zhou (2021) through a different surrogate-completion construction; we do not claim that the two estimators are identical or asymptotically equivalent.

Independently, D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021) developed *surrogate residuals* as a diagnostic device for discrete-choice models. Grounded in random utility theory (McFadden 1974), a surrogate maps an observed category to a draw from the conditional distribution of its latent utility. The result is a continuous, homoscedastic residual with a known reference distribution (Greenwell et al. 2018). We repurpose that construction as an *estimation* device. Substituting a surrogate S for the unobserved U turns the discrete problem into a continuous one on which the FWL recipe runs unchanged. The surrogate remains a draw rather than an observed outcome, so one completion is not enough. We instead treat repeated completions as multiple imputations and pool them with Rubin's rules (Rubin 1987). In the parametric designs that compare draw schemes, imputing from the model's *posterior predictive* distribution rather than a plug-in separates near-nominal coverage from clear under-coverage.

Contributions.

1. A DML-assisted estimator for binary and ordinal latent-utility effects in one framework, not tied to the logit link, with a built-in link diagnostic.
2. The finding that only the proper multiple-imputation draw scheme attained nominal coverage in the designs we study, quantified against naive and improper alternatives.
3. A closed-form analysis of how imputation-model error propagates to the effect, revealing a *first-order* sensitivity absent from standard DML, confirmed by a direct perturbation experiment.
4. Design-specific evidence for nominal coverage with machine-learning imputation models, using a partially-linear structure and a full-pipeline bootstrap, together with a proved deterministic-series reference learner.
5. An application to wine quality, illustrating an ordinal latent-scale estimand for which we are aware of no existing DML estimator.

2 Setup and estimand

Let D be a treatment, $X \in \mathbb{R}^p$ covariates, and Y a categorical outcome: binary, $Y = \mathbf{1}\{U > 0\}$, or ordinal, $Y = j \iff \kappa_{j-1} < U \leq \kappa_j$ for thresholds $-\infty = \kappa_0 < \kappa_1 < \cdots < \kappa_{J-1} < \kappa_J = \infty$ (we write κ rather than c to keep the thresholds distinct from the pass-through factor $c(\cdot)$ of Section 13). The latent utility follows a partially linear model

$$U = \theta_0 D + g_0(X) + \varepsilon, \qquad \varepsilon \mid D, X \sim F \tag{1}$$

with g_0 an arbitrary nuisance function and F a known error distribution (logistic, giving proportional-odds; extreme-value, giving complementary-log-log; etc.). The target θ_0 is the treatment effect on the latent-utility scale, the same quantity a correctly specified cumulative-link model targets, but with g_0 left nonparametric. It is not the probability-scale average treatment effect; the two answer different questions and should not be compared against a common truth. Because F fixes the variance of ε, this scale is set by the link rather than estimated, which is what gives θ_0 its familiar reading: under the logistic link a one-unit change in D shifts the cumulative log-odds of Y by θ_0, so $\exp(\theta_0)$ is a proportional-odds ratio.

Four assumptions identify θ_0.

1. *Selection on observables.* No unobserved confounder acts beyond X. Together with the conditional location-family statement in Equation 1, this assumption requires the latent error to have the same normalized law across treatment values given X.
2. *A constant latent-scale effect.* The partially linear form in Equation 1 makes θ_0 a single number rather than a function of X.
3. *A correctly specified latent distribution.* The completion draws from F using the assumed cutpoints, so the conditional law of ε, not merely its mean, must be correct. The link diagnostic monitors this assumption, and the propagation analysis shows that the estimate is especially sensitive to its failure.
4. *Overlap.* Treatment retains support after adjustment, so its residual is not annihilated.

The first and fourth assumptions are standard causal conditions, made explicit through the directed acyclic graphs (DAGs) for the simulations and application. The second and third are modeling choices shared with any cumulative-link analysis.

The inline example below and every simulation use the same causal DAG (Figure 1). Their functional forms, treatment type, error law, and cutpoints vary, but their causal arrows do not. The exogenous latent error affects the utility but is independent of treatment and covariates.

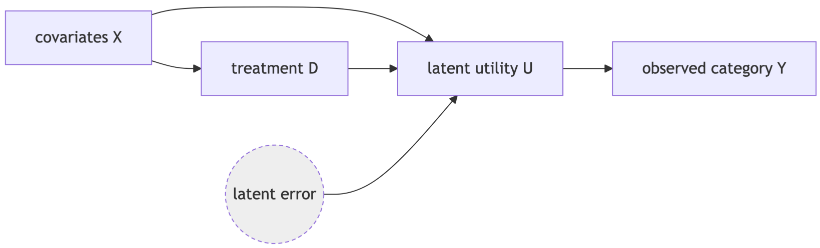

Figure 1: Causal DAG shared by the inline example and all simulation data-generating processes. The covariates may be a vector, and the observed category is a deterministic coarsening of the latent utility.

3 Background: FWL, DML, and surrogate residuals

The FWL theorem states that, in a linear model, the coefficient on D equals the coefficient from regressing the outcome residual on the treatment residual after partialling X out of both. DML makes this identity useful with high-dimensional covariates by combining flexible estimates of the two partialling regressions, K-fold cross-fitting, and an orthogonal score (Chernozhukov et al. 2018).

Applying that calculation directly to a binary or ordinal label targets a different quantity. A binary residual, for example, is bounded and heteroscedastic, so its regression on the treatment residual estimates a probability-scale contrast. SUDO instead uses the surrogate-residual construction (D. Liu and Zhang 2018; Cheng, Wang, and Zhang 2021) to complete the latent outcome before FWL is applied. The completion preserves the continuous latent scale needed to target θ_0.

4 Surrogate FWL: the idea in its simplest form

Fit an imputation model of the outcome on (D, X) to obtain an estimated latent index $\widehat{V}_i$. For each observation, draw a surrogate S_i from F, centered at $\widehat{V}_i$ and truncated to the interval implied by the observed category: $(0, \infty)$ when $Y_i = 1$, $(-\infty, 0]$ when $Y_i = 0$, or $(\hat{\kappa}_{Y_i-1}, \hat{\kappa}_{Y_i}]$ in the ordinal case. When the imputation model is correct and its parameters are known, this is the conditional law of U given (D, X). Therefore, $E[S \mid D, X] = E[U \mid D, X]$ (Theorem 1 of Cheng, Wang, and Zhang (2021)) and the completed data satisfy

$$S = \theta_0 D + g_0(X) + \eta, \qquad E[\eta \mid D, X] = 0 \tag{2}$$

with the same g_0 as in Equation 1, so FWL recovers θ_0. This is the exact case. With an estimated index $\widehat{V}$ the equality holds only asymptotically under consistent estimation, and approximately otherwise; the propagation analysis below makes the residual error precise. The demonstration that follows estimates the index by maximum likelihood, so it illustrates the corresponding large-sample regime.

The following self-contained example uses base R to draw a binary outcome from a logistic latent model, fit the imputation model, and complete the outcome to a surrogate. It compares three estimates of θ_0: the logistic-regression coefficient on the treatment, the FWL estimate on the *true* (unobserved) utility (the oracle), and the FWL estimate on the *surrogate* completion.

```
set.seed(1)
n <- 40000
theta <- 1.5
X <- rnorm(n)
D <- rbinom(n, 1, plogis(0.8 * X))  # treatment depends on a covariate
U <- theta * D + X + rlogis(n)      # latent utility, never observed
Y <- as.integer(U > 0)              # observed binary label

imputation_fit <- glm(Y ~ D + X, binomial)
index_hat <- predict(imputation_fit)
cdf_at_zero <- plogis(-index_hat)
```

```
uniform_draw <- ifelse(
  Y == 1,
  cdf_at_zero + runif(n) * (1 - cdf_at_zero),
  runif(n) * cdf_at_zero
)
surrogate <- index_hat + qlogis(
  pmin(pmax(uniform_draw, 1e-9), 1 - 1e-9)
)

treatment_residual <- resid(lm(D ~ X))
fwl <- function(outcome) {
  sum(treatment_residual * resid(lm(outcome ~ X))) /
    sum(treatment_residual^2)
}
round(c(
  glm = unname(coef(imputation_fit)["D"]),
  oracle = fwl(U),              # FWL on the true utility
  surrogate = fwl(surrogate)    # FWL on the completed utility
), 3)
```

```
      glm    oracle surrogate
    1.497     1.496     1.505
```

All three estimates are close to $\theta_0 = 1.5$. Using only the binary labels, the surrogate completion recovers the coefficient that FWL would estimate if the latent utility were observed. Replacing the binary index with an ordinal cumulative-link index and truncating to category-specific intervals gives the ordinal construction. The remainder of the paper adds two elements to this basic estimator: machine-learning partialling and inference that accounts for the uncertainty hidden by a single completion.

5 Surrogate-assisted double machine learning

The estimator. Cross-fit the two adjustment nuisances $\hat{m}(X) \approx E[D \mid X]$ and $\hat{\ell}(X) \approx E[S \mid X]$, and form the residual-on-residual estimate

$$\hat{\theta} = \frac{\sum_i (S_i - \hat{\ell}(X_i))(D_i - \hat{m}(X_i))}{\sum_i (D_i - \hat{m}(X_i))^2} \tag{3}$$

The $\hat{\ell}$ regression uses X only: the index carried D's signal into S, and partialling on X alone strips it back out so the residual regression isolates θ_0. The score $\varphi = (S - \ell(X) - \theta(D - m(X)))(D - m(X))$ is Neyman-orthogonal in (ℓ, m): its derivatives in both nuisance directions vanish by the tower property, so first-order error in $\hat{\ell}$ or $\hat{m}$ does not bias $\hat{\theta}$. This protection covers the two *adjustment* nuisances but not the imputation model that produced $\hat{V}$; the next section analyzes the difference.

Proper multiple-imputation inference. A single completion understates uncertainty, so we draw B surrogates and pool with Rubin's rules (Rubin 1987): the estimate is $\bar{\theta} = B^{-1} \sum_b \hat{\theta}^{(b)}$ and the total variance is $T = \bar{W} + (1 + 1/B)B_*$, the mean within-draw sandwich variance plus the inflated between-draw variance, with a Barnard-Rubin small-sample t reference (Barnard and Rubin 1999). What matters is *how* the draws differ. Each surrogate is centered at an estimated index $\hat{V} = x'\hat{\beta}$; that estimate carries sampling error, and reusing one fitted $\hat{\beta}$ across all draws (an *improper* draw) leaves the shared error invisible to the between-draw variance, so T comes up short. A *proper* draw redraws the imputation model's parameters, $\beta^{(b)} \sim N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$ (thresholds included in the ordinal case), before each completion, the large-sample normal form of Rubin's posterior-predictive requirement. Table 1 shows the three schemes moving coverage from 0.80 (naive) to 0.93 (improper) to 0.96-0.97 (proper).

T must account for three things that vary from draw to draw: the sampling noise in $\hat{\theta}$ once a completion is fixed (the within-draw $\bar{W}$), the surrogate randomness in S, and the uncertainty in the imputation model itself. Improper draws discard the last of these.

Under standard regularity, the normal parameter draw above describes the same large-sample limit as the sampling distribution of $\widehat{\beta}$, its flat-prior posterior, and a parametric bootstrap. A proper draw is therefore the large-sample form of a parametric bootstrap of the imputation model, obtained from its covariance rather than by refitting. This approach requires a model that reports a usable covariance, a requirement that fails for the black-box learners considered below.

Ordinal outcomes and the link. The ordinal calculation is unchanged once a cumulative-link model supplies the index and thresholds; because the thresholds are parameters too, the proper draw perturbs them jointly with the coefficients, which is why we fit with a model whose covariance covers them. The error distribution F remains an assumption. Under a correct link, the surrogate residuals $R = S - \widehat{V}$ are homoscedastic. A smooth of R^2 on X, computed with an independent draw stream, can therefore flag link misspecification through variance drift that mean-based checks miss.

6 How imputation-model error propagates

The orthogonality established above protects $\widehat{\theta}$ from error in the adjustment nuisances but says nothing about the imputation model, whose error is baked into the completed data S rather than differenced away. That channel distinguishes SUDO from standard DML. The imputation model, unlike the two adjustment nuisances, sits outside the orthogonality guarantee, so the method is DML-assisted rather than fully orthogonal; its overall score is not insensitive to every estimated component.

The argument requires the fitted index to be effectively free of an observation's own outcome. Two estimation strategies provide that separation. A *flexible* imputation model is cross-fit, making $\widehat{V}_i$ an out-of-fold prediction that is independent of Y_i given (D, X). A *parametric* model is fit once on the full sample; its $\widehat{V}_i$ depends on Y_i only at order $1/n$, which is asymptotically negligible. In our experiments, cross-fitting the parametric model was harmful: per-fold index noise inflated the between-draw variance, raising coverage to 1.00 while the mean standard error reached 2.3 times the true standard deviation. The binary machine-learning design therefore uses a cross-fit `gam` imputation model, whereas the ordinal designs and wine application fit `clm` once.

Either way $\widehat{V}$ is fixed with respect to the completion, while $Y \mid D, X$ is generated by the truth. Since S is $\widehat{V}$ plus truncated noise,

$$E[S \mid D, X] = \psi(\widehat{V}, \eta), \quad \psi(v, e) = v + F(e)\,\mu_+(v) + (1 - F(e))\,\mu_-(v) \tag{4}$$

with $\mu_\pm$ the truncated-noise mean corrections. The correctly centered draw is exact, $\psi(e, e) = e$ (recovering Theorem 1 of Cheng, Wang, and Zhang (2021)), but index error passes through with a factor strictly below one,

$$c(e) = \frac{\partial \psi}{\partial v}\bigg|_{v=e} = 1 - f(-e)\,(\mu_+(e) - \mu_-(e)), \qquad f = F' \tag{5}$$

Read c as the fraction of index error that survives the truncation. With no truncation the surrogate would be index plus noise and its mean would move one-for-one with $\widehat{V}$, giving $c = 1$; the observed category drags it back, and c is what is left.

Under a logistic link c rises from 0.31 at $e = 0$ to 1 in the tails (Figure 2). The value at the origin is exactly $1 - \log 2$, because for the logit

$$c(e) = 1 - H(F(e)), \qquad H(p) = -p \log p - (1 - p)\log(1 - p), \tag{6}$$

so the self-correction is precisely the entropy of the outcome. This identity summarizes the mechanism. At $p = 1/2$, the label is maximally uncertain, so observing it is most informative about where U fell and the correction is largest. As $p \to 1$, the label was already predictable, adds little information beyond $\widehat{V}$, and provides almost no correction. Strong effects are therefore difficult for the same reason that an ordinal category spanning many error standard deviations is difficult: in both settings, the observed category conveys little additional information about U.

The density in Equation 5 is evaluated at $-e$, which matters for an asymmetric F such as the extreme-value link; writing $f(e)$ is a shorthand valid only when F is symmetric. Section 13 gives the general-J form of both Equation 4 and Equation 5, where the ordinal thresholds contribute their own derivative block. For the well-placed partitions studied here, adding categories lowers the pass-through at $e = 0$: from 0.307 for binary to 0.157 and 0.090 for the displayed three- and four-category configurations. Category count alone is not monotone, however; widely spaced thresholds can instead drive the pass-through toward one.

Linearizing ψ around the truth, the bias of $\widehat{\theta}$ is approximately $\bar{c}_1\,\delta_D$ plus a covariate-direction leak given below, where δ_D is the imputation model's error in its effective treatment coefficient and $\bar{c}_1 = E[w\,c(\theta_0 + g)]/E[w]$, $w = m(1-m)$.

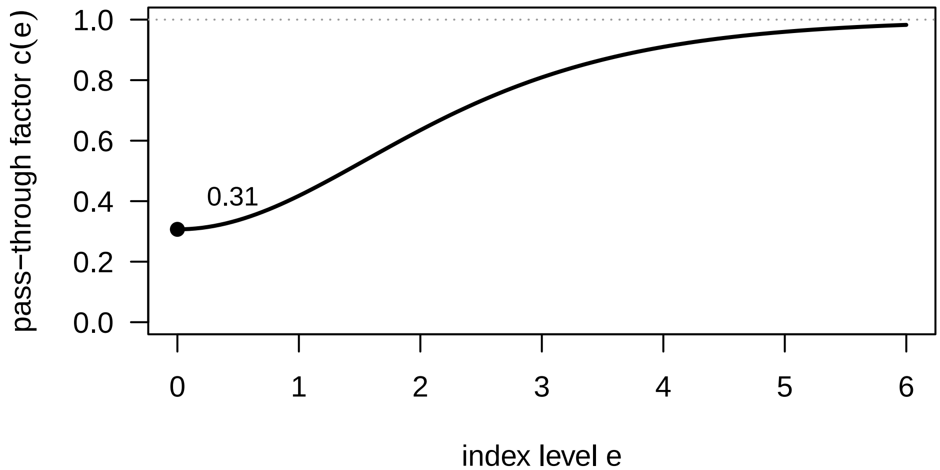

Figure 2: Pass-through factor $c(e)$ under a logistic link. Error in the imputation index reaches $\hat{\theta}$ multiplied by $c(e)$. The factor rises from about 0.31 at $e = 0$ toward 1 as the signal grows because truncation self-corrects when Y is informative but not when Y is nearly deterministic. Averaging over the index distribution gives the $\bar{c}_1$ reported in the text.

Self-correction weakens with signal. Because Y comes from the truth, truncation pulls S toward the correct interval, giving $c < 1$. As the signal grows and Y becomes nearly deterministic, however, it adds no information beyond $\hat{V}$ and $c \to 1$. This loss of self-correction, rather than numerical instability, makes inference hardest at strong effect sizes.

Covariate-direction error can survive partialling. The opposite might seem natural because residualized treatment is orthogonal to any function of X. To see why it fails here, split the imputation model's index error into a treatment part and a covariate part, $\hat{V} - \eta = \delta_D D + \delta_X(X)$. Partialling-out regresses on $R = D - m_0(X)$, and $E[R \mid X] = 0$ annihilates any function of X, so δ_X ought to wash out. It does not, because S is nonlinear in $\hat{V}$: linearizing multiplies the index error by $c(\eta)$, and η depends on D, so δ_X arrives multiplied by something that itself varies with treatment and therefore survives the projection. For binary D, writing $\eta_d = \theta_0 d + g_0(X)$ and $w = \mathrm{Var}(D \mid X)$, the surviving term is

$$\text{covariate-direction leak} \;=\; \frac{E[w\,(c(\eta_1) - c(\eta_0))\,\delta_X]}{E[w]}. \tag{7}$$

Proposition A4 gives the exact path-integrated version of this first-order form and a weighted norm bound. The leak is governed by *how differently the truncation damps the two treatment arms*. Were the two path-averaged factors the same, it would vanish exactly and the naive argument would be right.

The leak vanishes when the imputation model can represent g_0, which is why designs with every model correctly specified show no covariate-direction bias. When it cannot, two channels can appear. Giving g_0 an X_1X_2 interaction that an additive logistic imputation model cannot express produces a bias of 5 to 6% of θ_0 and coverage of 0.75 at strong signal. Although treatment is constrained to enter linearly, omitted logistic heterogeneity shifts its pseudo-true coefficient, so this experiment is not a pure $\delta_D = 0$ case. The population decomposition in Section 15 separates the common covariate leak from that induced treatment-direction error. Refitting the same design with an imputation model that admits the interaction removes the bias, confirming the imputation model rather than the adjustment nuisances as the source.

Imputation-model error is first order. In the sharpest contrast with standard DML, the imputation model's error enters $\hat{\theta}$ *linearly*. In DML the outcome is observed and nuisance error enters only through a *product* of two errors (Chernozhukov et al. 2018; Robinson 1988): writing $\ell = \ell_0 + a$ and $m = m_0 + b$ for the errors in the two adjustment models, the remaining bias is $E[ab] - \theta_0 E[b^2]$, which is second order in the pair. Getting the treatment model right ($b = 0$) removes it entirely; getting only the outcome model right does not. The score is Neyman-orthogonal, first-order insensitive to nuisance error, rather than doubly robust in the usual sense. SUDO adds a third model whose treatment-direction error is unprotected and first-order, damped only by $\bar{c}_1 < 1$. Table 3 confirms this numerically: a perturbation applied to the outcome nuisance in DML leaves the estimate unbiased at every level, whereas the same-sized index perturbation in SUDO produces a linear bias with slope exactly $\bar{c}_1$.

7 Machine-learning imputation models

Equation 4 reverses the usual DML emphasis: imputation-model quality matters more than adjustment-model quality. A well-specified parametric imputation model (logistic or cumulative-link) needs no extra machinery, since its covariance supplies the proper draw directly. Such draws propagate parameter uncertainty and attain near-nominal coverage in our parametric experiments; Section 13 characterizes the small Rubin-variance discrepancy that remains. A black-box learner introduces two obstacles: it can carry treatment-direction error δ_D, creating bias, and it has no covariance matrix from which to draw, complicating inference.

The bias is addressed by a *partially-linear* (PL) imputation model, $\hat{V} = \hat{\alpha} D + \hat{f}(X)$, in which the treatment enters linearly and machine learning handles only X, so no treatment-by-covariate artifact can arise. This restriction blocks one important source of treatment-direction error, but it does not guarantee that $\hat{f}$ is accurate or that the learner satisfies the projected expansion needed for inference. It defines a structural learner class, not a commitment to one training algorithm. We implement it with a generalized additive model (GAM), multivariate adaptive regression spline (MARS) bases followed by a joint logistic fit, and neural-network backfitting. All three adaptive learners meet the Monte Carlo-aware bias criterion across effect sizes in the common comparison in Section 8. We also implement a deterministic Hermite-series learner as a proof reference. Theorem S1 proves its projected first-order expansion, and Theorem S2 proves bootstrap stability under explicit series conditions; it is not the practical default.

For inference, we use a *full-pipeline bootstrap*, the nonparametric counterpart of the proper draw. A parametric model injects imputation-model uncertainty by sampling β from its covariance. Here we instead resample the whole data set and rerun the entire procedure, cross-fitting and all, using the spread of the estimates as the standard error. Both approaches target the same variance component; the bootstrap obtains it by refitting when no covariance is available. It has to wrap the *whole* pipeline, because a within-fold refit of the model alone under-propagates that same imputation-model variability. In the designs studied, normal intervals using this full-pipeline bootstrap attain nominal coverage for all three adaptive learners (Section 8).

Conjecture M2 states a learner-class bootstrap result under a projected bootstrap-stability condition and conditional nuisance rates. Under those conditions, fixed and redrawn regular folds are both admissible, and only the integrated generated-outcome map must be differentiable, not every raw inverse-transform draw. The conjecture also identifies a fixed-imputation centering issue. Fresh surrogate randomness makes the bootstrap variance consistent, but it does not justify the usual distribution centered at one realized randomized estimate. The supported procedure is therefore the normal interval based on the full-pipeline bootstrap standard deviation.

Learner-specific verification of the projected stability condition remains open for the three adaptive algorithms. The deterministic series result closes the proof route for one tractable partially-linear learner; the adaptive-learner comparison below supports, but does not prove, the same conclusion for GAM, MARS, or neural backfitting.

A Bayesian ensemble offers a direct alternative because each posterior draw is a complete imputation model and therefore a proper draw. Under a probit imputation model, the Albert-Chib augmentation variable is the surrogate completion, so each Gibbs sweep supplies one. In the probit analogue we studied, however, the approach did not improve performance. Bayesian additive regression tree (BART) imputation, both unconstrained and under the partially-linear restriction above, had bias that grew with signal strength to about 5% of θ_0, where the backfit meets the bias criterion. Its posterior draws propagated variance no better than the bootstrap they were intended to replace. This result comes from one design and is not a general verdict. The run is in the repository. A forest's out-of-bag or infinitesimal-jackknife variance (Wager, Hastie, and Efron 2014) is not such a route at all: it quantifies prediction uncertainty at a point, not a coherent joint draw of the whole imputation model.

8 Simulation study

Every simulation data-generating process (DGP) follows the causal structure in Figure 1. Independently and identically distributed (iid) covariates determine treatment assignment; treatment and covariates then determine a latent utility, which is coarsened into the observed category:

$$X_1, \dots, X_p \overset{\text{iid}}{\sim} N(0,1), \qquad D \mid X \sim \text{Bernoulli}(\Lambda(\pi(X))), \qquad U = \theta_0 D + g_0(X) + \varepsilon, \tag{8}$$
$$\varepsilon \sim F, \qquad Y = j \iff \kappa_{j-1} < U \le \kappa_j,$$

where Λ is the logistic cumulative distribution function. Choosing $(p, \pi, g_0, F, \kappa, \theta_0, n)$ then fixes a design. We use two covariate structures throughout. The **one-covariate setting (S1)** has $\pi(X) = 0.8X$ and $g_0(X) = X$; every fitted model is correctly specified, so only pooling is under test. The **five-covariate machine-learning setting (S5)** has

$$\pi(X) = X_4 + \cos X_5, \qquad g_0(X) = X_1^2 + \sin X_2 + 0.5X_3,$$

which is nonlinear in X and makes the treatment depend on covariates that do not enter g_0. Binary outcomes take $\kappa = (0)$; ordinal ones cut U at two interior thresholds into $J = 3$ categories. Table 8 in the appendix lists every design's parameters.

The coverage target is 0.95, and Monte Carlo error for a reported coverage rate is about ± 0.02. **Standard deviation (SD)** denotes the estimator's run-to-run variability; **mean standard error (mean SE)** is the average uncertainty it reports. A mean SE below the SD signals intervals that are too short. Section 14 gives learner and tuning specifications, fold and draw counts, and per-cell Monte Carlo standard errors. The tables below are generated at render time from the committed result files.

Proper draws are what deliver coverage. With a simple binary design and a correct parametric imputation model, only the proper scheme reaches nominal coverage; the naive interval covers 0.80 and the improper 0.93 (Table 1).

Table 1: Inference schemes, binary outcome, $n = 5000$, $\theta_0 = 1.5$, 500 replications. B is the number of surrogate draws.

Scheme	Bias	SD	Mean SE	Coverage
Naive single draw	+0.003	0.083	0.055	0.804
Improper, $B = 25$	+0.004	0.071	0.067	0.924
Proper, $B = 25$	+0.005	0.072	0.076	0.970

Ordinal outcomes. With a three-category outcome and machine-learning adjustment, SUDO meets its bias criterion with valid-to-conservative coverage (Table 2), a regime for which we are aware of no DML competitor for the latent-scale coefficient. The original implementation was substantially conservative. A term-level ordinal decomposition includes the threshold block and matches Monte Carlo throughout a clean cumulative-logit testbed. The decomposition predicts only 2.8% variance overstatement at moderate signal and 2.0% understatement at strong signal, so threshold estimation and ordinal pooling alone cannot explain the 50% to 88% variance gap after flexible partialling nuisances are introduced. A paired nuisance-isolation experiment locates the gap in the fixed outcome nuisance: refitting $E[S \mid X]$ for every ordinal completion reduces mean T by 38% while leaving the empirical sampling variance unchanged, moving T/V from 1.505 to 0.942. The full corrected stage at $B = 25$ and 200 replications has bias 0.016, coverage 0.960, and $T/V = 1.214$ with Monte Carlo standard error 0.122. Per-draw refitting is therefore the validated ordinal procedure, at the cost of fitting the outcome nuisance B times rather than once. The residual variance ratio is compatible with one at this simulation's precision and is not evidence of a remaining ordinal anomaly. Rubin's variance is guaranteed to err safe only under conditions we have not verified here (a valid imputation model together with a self-efficient complete-data analysis), and under uncongeniality it can be biased in either direction (Meng 1994; Xie and Meng 2017). The over-coverage is a property of these designs, not a general guarantee.

Table 2: Ordinal outcome, $J = 3$, $\theta_0 = 1.0$. Parametric arm 500 replications; corrected spline + ML arm 200, with the outcome nuisance refit per completion.

Configuration	Bias	SD	Mean SE	Coverage
Parametric, $n = 3000$	+0.002	0.077	0.083	0.976
Spline + ML nuisances, $n = 2000$	+0.016	0.104	0.114	0.960

Link misspecification. Generating Gumbel noise but assuming a logistic link substantially biases the latent-scale effect (+0.77 for binary, +0.58 for ordinal outcomes, with zero coverage). The correct link removes this bias (bias < 0.005, nominal coverage). The variance-drift diagnostic flags the wrong link ($p \approx 2 \times 10^{-4}$) and clears the correct one ($p \approx 0.43$).

First-order versus second-order sensitivity. Table 3 traces the bias after perturbing one input by a known function. Standard DML with the observed outcome remains unbiased under an arbitrarily wrong outcome nuisance when the propensity is correct. SUDO's index error instead enters linearly: its treatment-direction slope matches the derived $\bar{c}_1$ (0.669 at $\theta_0 = 1.5$, 0.861 at $\theta_0 = 3$) to within 0.001.

Table 3: Nuisance-sensitivity ladder.

Arm	Perturbed	Bias vs. perturbation
DML	outcome nuisance only	flat, ≈ 0

Arm	Perturbed	Bias vs. perturbation
DML	both nuisances	quadratic
SUDO	index, treatment direction	linear, slope $= \bar{c}_1$

Partially-linear imputation learners. A tuned partially-linear backfit meets the bias criterion at both effect sizes. Pairing it with the full-pipeline bootstrap brings strong-effect normal coverage from 0.86 to 0.98; the corresponding percentile coverage is 0.95. The normal interval is the inferential procedure supported by Conjecture M2 under its stated conditions. The percentile result is retained as an empirical comparison from the original validation. With 100 replications, either coverage estimate has Monte Carlo uncertainty of about ± 0.03; the bootstrap SD also matches the estimator's run-to-run variability at this resolution.

Backfitting is not essential. A common-sample comparison fits the same $\alpha D + f(X)$ structure three ways: a GAM with a linear treatment term, MARS bases constructed from X only followed by a joint logistic fit with mandatory linear D, and the tuned neural-network backfit. The MARS configuration is selected on theta-level Monte Carlo bias, not predictive generalized cross-validation (GCV). The three adaptive learners pass the original prespecified bias and percentile-coverage criteria (Table 4). Their normal coverage, which is the recommended inferential comparison after the fixed-B centering analysis, ranges from 0.94 to 0.98. With 50 replications, these coverage cells have Monte Carlo standard errors of roughly 0.02 to 0.05, so their small differences do not rank the learners.

Table 4: Partially-linear imputation learners with full-pipeline bootstrap, $n = 2000$, 50 replications, 50 outer resamples, and 15 surrogate completions per resample.

Learner	θ_0	Bias	SD/Mean SE	Normal coverage	Percentile coverage
GAM	1.5	+0.010	0.948	0.980	0.940
GAM	3.0	+0.043	0.960	0.940	0.880
MARS	1.5	-0.009	0.978	0.960	0.920
MARS	3.0	-0.010	0.945	0.960	0.940
Neural backfit	1.5	-0.005	0.978	0.940	0.940
Neural backfit	3.0	+0.007	0.998	0.980	0.960

The bootstrap comparison uses paired generated datasets and reruns each learner with either one fixed regular fold partition or folds redrawn in every resample. Table 5 reports the direct comparison. The within-fold method refits only the imputation learner inside the original training folds, while the improper method holds its fitted index fixed. Their SE ratios therefore locate which layer of variation is omitted. The normal interval is the procedure supported by Conjecture M2 under its high-level conditions; percentile and conventionally centered studentized intervals are included as fixed-B empirical benchmarks.

Table 5: Paired full-pipeline bootstrap comparison, $n = 2000$, 30 replications, 15 outer resamples, and 15 surrogate completions per resample.

Learner	θ_0	Fixed / redrawn SE	Within / full SE	Improper / within SE	Normal coverage	Percentile coverage	Studentized coverage
GAM	1.5	0.971	0.896	0.875	0.967	0.900	0.900
MARS	1.5	0.966	0.935	0.873	0.967	0.900	0.933
Neural backfit	1.5	1.038	0.928	0.921	0.933	0.867	0.800
GAM	3.0	0.983	0.754	0.753	0.967	0.933	0.933
MARS	3.0	0.976	0.769	0.761	0.967	0.967	0.867
Neural backfit	3.0	1.079	0.768	0.809	0.967	0.867	0.900

Continuous treatment, and other variations. The designs above all use a binary treatment, while the application below does not, so we check the transfer directly. Replacing binary D with $D = 0.7\pi(X) + N(0, 1)$ in the S5 structure leaves the estimator unbiased for both a binary outcome (-0.1% at $\theta_0 = 1$, $+0.4\%$ at 2.5) and a three-category ordinal one ($+0.7\%$ and $+1.5\%$), so the latent-scale estimand and the machinery carry over. Doubling the covariate count to ten, and scaling the treatment-assignment index so that overlap is markedly worse, likewise cost nothing in bias. Bias does appear when the imputation model cannot represent g_0, as described above. Coverage under a continuous treatment tracks the binary case at moderate signal but is worse at strong signal, where the reported SE falls furthest short of the truth. The full-pipeline bootstrap is the remedy there, as it is for the black-box case. Details and the full grid are in Table 8 and the repository.

9 Application: volatile acidity and wine quality

We estimate the effect of volatile acidity, the acetic-acid fault commonly described as a vinegar or nail-polish aroma, on wine quality. In the University of California, Irvine (UCI) wine-quality data (Cortez et al. 2009), quality is the median ordinal score from blind tasters. The data contain 1,599 red and 4,898 white wines, with quality spanning six to seven levels.

Because these data are observational, causal interpretation depends on the assumed application DAG in Figure 3. Volatile acidity is a plausible treatment: tasters are expected to penalize the acetic fault, and the variable is a fault endpoint rather than an apparent cause of the remaining chemistry. Even so, the interpretation requires the other physicochemical measurements to be pre-treatment adjustments rather than mediators or proxies of volatile acidity. It also requires winemaking practice to have no direct effect on quality beyond measured chemistry, the exclusion restriction encoded by the graph.

We do not claim that these assumptions are established. Instead, we examine the two most questionable features through adjustment-set robustness and an unmeasured-confounding calibration. The application illustrates the method on a genuinely ordinal outcome; it is not a settled causal finding.

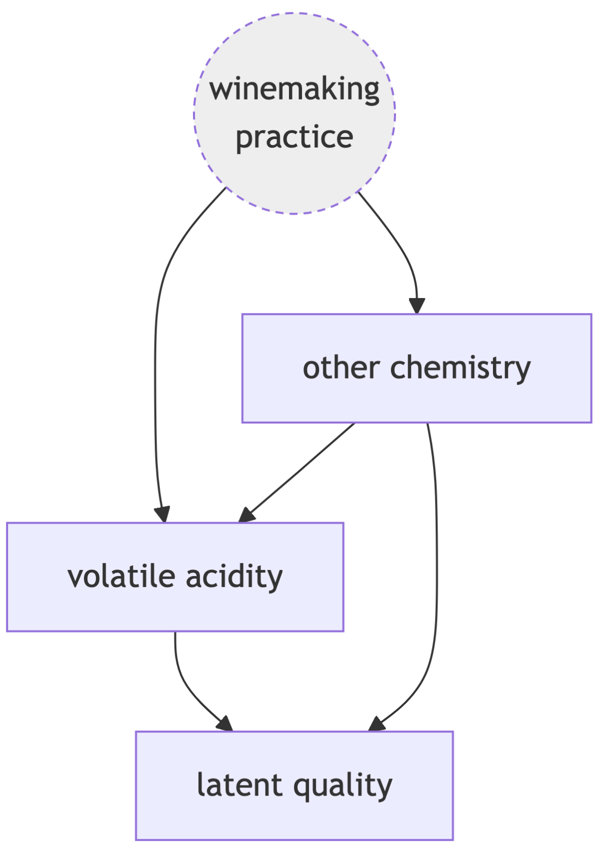

Figure 3: Assumed application DAG. Volatile acidity (the treatment) affects latent quality directly; unobserved winemaking practice confounds both through the measured chemistry, the adjustment set.

Volatile acidity retains adequate residual variation after adjustment: 75% of its standard deviation for red wine and 94% for white wine. These values are residual $\sqrt{1 - R^2}$ computed from the R^2 column of Table 6. The treatment residual is therefore not annihilated, and the residual-on-residual regression is well posed.

Ordinal SUDO, using a cumulative-link imputation model, proper draws, cross-fitted nuisances, and Rubin pooling, gives a negative effect on latent quality. The estimate is stable across linear and partially-linear spline imputation models (Table 6). As a model-based comparator, we also report the treatment coefficient from the cumulative-link fit itself. Its standard error is smaller (white: 0.031 versus 0.053-0.074) because it relies directly on the full parametric outcome model, whereas SUDO leaves the partialling regressions flexible and propagates the completion and cross-fitting variability. The narrower model-based interval can be efficient when its stronger functional-form assumptions are correct; it is not evidence that SUDO's additional uncertainty is spurious.

Volatile acidity is standardized before fitting, so these coefficients are per one standard deviation (about 0.18 g/dm^3 for red, 0.10 for white), which keeps the effect comparable across the two wines instead of tied to the raw acetic-acid scale. The proportional-odds reading then makes the magnitude concrete: a one-SD rise in volatile acidity multiplies the odds of a wine scoring above any fixed quality level by $\exp(\hat{\theta})$, about 0.59 for red ($\hat{\theta} = -0.53$) and 0.62 to 0.66 for white, a reduction of roughly a third to 40% in those odds per standard deviation.

Table 6: Effect of a one-SD increase in volatile acidity on latent wine quality. Model-based coefficient from the linear cumulative-link fit (standard error in parentheses); SUDO estimates with 95% intervals in brackets.

Wine	Overlap $R^2(D \sim X)$	Cumulative-link	SUDO (linear)	SUDO (spline)
Red	0.44	-0.61 (0.07)	-0.53 [$-0.68, -0.38$]	-0.53 [$-0.69, -0.36$]
White	0.12	-0.50 (0.03)	-0.41 [$-0.51, -0.31$]	-0.48 [$-0.63, -0.34$]

The negative sign is stable across the reported sensitivity checks. Dropping any single covariate from the adjustment set leaves the effect in $[-0.67, -0.46]$ for red and $[-0.49, -0.37]$ for white, every interval excluding zero, so the estimate does not hinge on one adjustment choice. Read through the omitted-variable-bias framework (Cinelli and Hazlett 2020), an unobserved confounder would have to explain about 16% (red) and 11% (white) of the residual variation in both volatile acidity and latent quality to bring the effect to zero; weaker confounding leaves it negative. We report these robustness values as a heuristic calibration rather than a formal bound: the Cinelli-Hazlett formula is derived for omitted-variable bias in linear regression with an observed outcome, and we are applying it to a pooled, surrogate-completed estimate whose standard error comes from Rubin's rules. Its validity in that setting has not been established. These values and the leave-one-out ranges are in the committed result files.

How wrong could the imputation model be? Unmeasured confounding is the threat any observational analysis faces; the one specific to SUDO is error in the imputation model, which Equation 7 and Equation 5 show reaches $\hat{\theta}$ at first order. That channel admits its own sensitivity statement, and unlike the robustness value above it follows directly from the estimator's own theory rather than being imported. Since the bias is $\bar{c}_1 \delta_D$, the imputation model's error in the volatile-acidity coefficient that would drive the estimate to zero is $\delta_D = -\hat{\theta}/\bar{c}_1$. Evaluating $\bar{c}_1$ at the fitted index with the continuous-treatment multiplier $E[c(\eta)DR]/E[R^2]$ and the general-J pass-through of Proposition A1 gives $\bar{c}_1 = 0.21$ (red) and 0.17 (white). The cumulative-link model would therefore have to be wrong about volatile acidity by $+2.55$ and $+2.38$ respectively, which is 4.2 and 4.8 times the coefficient it actually fits. This calculation addresses treatment-direction error only. Conditional on covariate-direction misspecification being negligible, it shows that a large treatment-coefficient error would be required to erase the estimate; it does not establish robustness to general imputation-model error.

Two features of the design produce this result. First, the pass-through is near 0.2, rather than the 0.67 to 0.86 found in the binary simulations. Six or seven categories with well-spaced thresholds constrain the latent utility tightly, allowing truncation to provide substantial self-correction. Second, the estimate is far from zero relative to its scale. Both protections weaken for a coarser outcome or a smaller effect, precisely the settings in which this sensitivity calculation becomes most informative.

10 Discussion

SUDO uses outcome completion to extend partially linear DML to binary and, to our knowledge for the first time, ordinal latent-scale effects. Other DML scores can target probability- or distribution-scale effects for categorical outcomes; SUDO targets the coefficient of a latent-utility model. Under the logit link it targets the same latent-scale coefficient as the binary-logistic DML estimator of M. Liu, Zhang, and Zhou (2021), while admitting other error distributions and ordinal as well as binary responses. The methodological core is inferential: the completions are multiple imputations, and in our experiments only proper draws, which redraw the imputation model's parameters (thresholds included), attain nominal coverage. The method is link-flexible over any distribution for which the truncated draw and reference are available, and its residuals give a diagnostic that can flag, though not confirm, a wrong link.

The propagation analysis makes the main tradeoff explicit. The imputation model sits outside the orthogonality guarantee, so its error reaches the effect at first order through a truncation factor that increases as the signal strengthens and the outcome's self-correction fades. Standard DML nuisance error is instead second order. In the designs studied, this sensitivity is manageable: well-specified parametric models with proper draws and the three validated partially-linear learners with a full-pipeline bootstrap attain nominal coverage. Partial linearity alone does not provide this guarantee. The imputation learner must still estimate the covariate component well and satisfy the projected stability conditions needed for inference.

Several problems remain open. Section 13 separates the proved results from the conjectured ones. Theorem S1 proves the projected expansion for a deterministic logistic-series learner, and Theorem S2 verifies its full-pipeline bootstrap stability under explicit series and conditional bootstrap conditions. The broader adaptive-learner statements remain Conjectures M1 and M2: their regularity conditions are stated but not minimized, and the projected asymptotic-linearity condition has not been proved separately for GAM, MARS, or neural backfitting. SUDO therefore has a proved projected expansion and bootstrap result for the theorem-reference specialization, while inference with the adaptive learners remains simulation-supported. Natural next steps are to extend the adaptive results beyond the data-generating structures studied and to develop a method for unordered (multinomial) outcomes, which require a vector surrogate and have no single scalar effect. A Bayesian ensemble could replace the bootstrap because its posterior draws are literal proper imputations. A BART version did not improve on the bootstrap in the one design we tried, so the conditions under which that route succeeds remain open.

11 Data and software

The wine-quality data come from the University of California, Irvine Machine Learning Repository (Cortez et al. 2009). The `sudo` Python package provides the estimator, and a parallel set of R scripts prototypes and validates each step. The open-source project repository contains the code and committed result files behind every table and figure under the Massachusetts Institute of Technology (MIT) License: https://github.com/bgreenwell/sudo.

12 Declaration of artificial intelligence use

We used generative artificial intelligence (AI) to assist with code and summaries of reference materials. The authors reviewed and validated all outputs, retain full ownership of the final manuscript, and accept sole responsibility for any errors.

13 Appendix: asymptotic theory

This appendix develops four theoretical components used in the body: the pass-through map and its derivatives for general J; the moment properties that separate adjustment nuisances from the imputation model; the fixed-B asymptotic distribution of the pooled estimate; and the resulting account of Rubin's variance.

The labels distinguish the status of each result. Direct differentiation, conditioning, or inequalities establish the propositions. The general asymptotic expansions and adaptive-learner bootstrap statements are labeled conjectures because the manuscript gives heuristic derivations and numerical verification, not complete proofs. The deterministic-series specialization is labeled a theorem because complete proofs are supplied. The learner-specific projected asymptotic-linearity and bootstrap-stability conditions remain unproved for the implemented adaptive learners. The project repository checks every closed-form expression numerically, using numerical differentiation for the pass-through derivatives and Monte Carlo for the variance decomposition.

13.1 Notation and assumptions

Let the imputation model be a parametric family $P_\beta(Y \mid D, X)$ with latent index $\eta(D, X; \beta)$. For a generalized linear model (GLM) with known cutpoints, $\eta = x'\beta$. When a cumulative-link model estimates the cutpoints, we collect them and the index coefficients in $\beta = (\beta_\eta, \kappa)$. The thresholds are components of β, but not of η; they enter only through the truncation interval. This distinction matters in Equation 11. Write $\hat{\beta}$ for the maximum-likelihood estimate, $m_0(X) = E[D \mid X]$, $R = D - m_0(X)$, and $J_\theta = E[R^2]$.

Given β, a completion is $S(W; \beta) = \eta(D, X; \beta) + \xi$ with ξ drawn from F truncated to the interval Y implies. The draw noise splits into a data-measurable part and a pure Monte Carlo part,

$$\xi_i^{(b)} = e_i + u_i^{(b)}, \qquad e_i = E[\xi_i \mid W_i; \beta], \qquad E[u_i^{(b)} \mid W_i] = 0, \tag{9}$$

This split determines the fixed-B variance decomposition. Note that ξ is *not* mean-zero given W: the truncation is selected by the observed Y. Only after averaging over $Y \mid D, X$ at the true model does the mean vanish, because composing $Y \sim P_{\beta_0}$ with the matching truncated draw reproduces the untruncated law, so $\xi \mid D, X \sim F$ exactly. Define

$$\sigma_e^2 = E[e^2 R^2], \quad \sigma_u^2 = E[\mathrm{Var}(\xi \mid W) R^2], \quad \sigma_\varphi^2 = E[\xi^2 R^2] = \sigma_e^2 + \sigma_u^2.$$

We assume throughout: **(A1)** the latent model Equation 1 with selection-on-observables, a constant latent-scale effect, conditional error law $\varepsilon \mid D, X \sim F$, and overlap ($J_\theta > 0$, $\mathrm{Var}(D \mid X)$ bounded away from 0 and ∞); **(A2)** a correctly specified, regular imputation model, so $\eta(D, X; \beta_0) = \theta_0 D + g_0(X)$ and $\sqrt{n}(\hat{\beta} - \beta_0) \to_d N(0, \Sigma_\beta)$ with $\Sigma_\beta = I(\beta_0)^{-1}$ the inverse per-observation information and influence function $\mathrm{IF}_\beta = I(\beta_0)^{-1} s_\beta(W)$; **(A3)** cross-fitted adjustment nuisances consistent in L_2 with the product rate $\|\hat{\ell} - \ell_0\|_2 \|\hat{m} - m_0\|_2 = o_p(n^{-1/2})$, each $o_p(n^{-1/4})$; **(A4)** finite fourth moments and dominated differentiability of $\beta \mapsto E[\varphi]$ at β_0; and **(A5)** proper draws $\beta^{(b)} \sim N(\hat{\beta}, \widehat{\Sigma}_\beta / n)$, so that $\beta^{(b)} = \hat{\beta} + n^{-1/2} Z_b$ with $Z_b \mid \text{data} \sim N(0, \Sigma_\beta)$.

The scaling in (A5) must be explicit because Σ_β is the limit of $n\widehat{\mathrm{Cov}}(\hat{\beta})$ rather than of $\widehat{\mathrm{Cov}}(\hat{\beta})$ itself; drawing with the unnormalized matrix would inject variance that does not vanish with n.

Scope. Assumption (A2) describes the parametric case: one regular maximum-likelihood fit on the full data, with an estimated covariance from which to draw. The ordinal designs and wine application use this case. Conjecture M1 separately treats a cross-fitted flexible imputation index and does not require a single $\hat{\beta}$ or covariance matrix. Its central condition is an asymptotically linear, residual-weighted projection of the fitted index error. This learner-level condition does not follow from cross-fitting or structural partial linearity alone. Theorems S1 and S2 establish it for the deterministic-series reference. For the three adaptive learners, we verify the generated-outcome linearization but do not claim a proof of their projected functional condition.

13.2 The pass-through map for general J

For an outcome in category j the draw truncates ξ to $(a_j, b_j]$ with $a_j = \kappa_{j-1} - v$ and $b_j = \kappa_j - v$ at index v. Write $\mu_j(v, \kappa) = E[\xi \mid a_j < \xi \le b_j]$ and $p_j = F(\kappa_j^0 - e) - F(\kappa_{j-1}^0 - e)$ for the true category probability at true index e. Averaging over $Y \mid D, X$ gives the general form of Equation 4,

$$\psi(v, \kappa; e, \kappa^0) = v + \sum_{j=1}^{J} p_j \, \mu_j(v, \kappa). \tag{10}$$

Proposition A1. *At $(v, \kappa) = (e, \kappa^0)$, writing $t_k = \kappa_k^0 - e$,*

$$c(e) = \frac{\partial \psi}{\partial v} = 1 - \sum_{k=1}^{J-1} f(t_k)[\mu_{k+1}(e) - \mu_k(e)], \qquad \frac{\partial \psi}{\partial \kappa_k} = f(t_k)[\mu_{k+1}(e) - \mu_k(e)]. \tag{11}$$

Consequently $c(e) = 1 - \sum_k \partial\psi/\partial\kappa_k$: the index derivative and the threshold derivatives are the same kernel.

Proof. Differentiate Equation 10. Both truncation limits move with slope -1 in v, and the derivative of a truncated mean with respect to its upper limit is $f(b)(b - \mu)/P$, with the analogous expression at the lower limit. Evaluated at the truth the truncation probability P_j equals p_j, so the weights in Equation 10 cancel the denominators and the surviving terms regroup by threshold. The final identity follows because μ_j depends on (v, κ) only through $\kappa - v$, so shifting the index is the same as shifting every threshold the other way. □

Proposition A1 has three practical implications. For binary outcomes, $J = 2$ and $\kappa_1 = 0$, so the sum reduces to Equation 5. For ordinal outcomes, the generally nonzero threshold derivatives require both Σ_β and the gradient G below to contain a threshold block. This requirement is why the ordinal full model must use a routine whose covariance covers the thresholds (`ordinal::clm`) rather than one whose covariance does not (`mgcv::ocat`).

The final implication concerns interpretation. Pass-through depends on how much the categorization reveals about U, not simply on the number of categories. Refining a partition with well-placed thresholds lowers c at $e = 0$ under a logit, from 0.307 for binary outcomes to 0.157 and 0.090 for the three- and four-category configurations in Section 15. Widening the same three-category partition reverses this pattern, raising $c(0)$ above the binary value and toward 1. The reason is visible in Equation 11: a category spanning many error standard deviations has a nearly unbounded truncation interval, so its μ_j barely moves with the index and self-correction switches off. We therefore make no monotonicity claim in J. The relevant limit is $c \to 1$ as the observed category ceases to constrain U, the same mechanism that operates at strong signal.

The binary logit closed form. For $J = 2$ under a logistic F, the truncated means have an exact expression that turns Equation 5 into Equation 6. Substituting $u = F(x)$, so that $x = \log\{u/(1-u)\}$ and $\mathrm{d}u = f(x)\,\mathrm{d}x$, the partial mean is

$$\int_{-\infty}^{t} x f(x)\,\mathrm{d}x = \int_{0}^{F(t)} \log \frac{u}{1-u}\,\mathrm{d}u = q\log q + (1-q)\log(1-q) = -H(q), \qquad q = F(t),$$

so the partial mean of a logistic is minus the entropy at the corresponding probability. Taking $t = -e$ and $p = F(e)$, and using $H(1-p) = H(p)$ with the fact that the untruncated mean is zero,

$$\mu_-(e) = \frac{-H(p)}{1-p}, \qquad \mu_+(e) = \frac{H(p)}{p}, \qquad \mu_+ - \mu_- = \frac{H(p)}{p(1-p)}.$$

For the logistic law, $f(-e) = f(e) = p(1-p)$, so this factor cancels exactly in Equation 5. Thus $c(e) = 1 - H(p)$ and $c(0) = 1 - \log 2 = 0.307$. The cancellation makes self-correction equal to the outcome entropy rather than merely proportional to it.

13.3 Cross-fitted partially linear imputation learners

The machine-learning estimator does not have a finite-dimensional β. Partition the observations into fixed-K folds I_k. For $i \in I_k$, fit the following index on I_k^c:

$$\hat{v}_{-k}(D, X) = \hat{\alpha}_{-k} D + \hat{f}_{-k}(X),$$

and form B completions using independent auxiliary uniforms. Let $v_0(D, X) = \theta_0 D + g_0(X)$ and define the conditional generated-outcome error

$$\Delta_k(D, X) = \psi\{\hat{v}_{-k}(D, X); v_0(D, X)\} - v_0(D, X). \tag{12}$$

For ordinal outcomes, the thresholds are additional arguments of ψ. The scalar-index result below applies directly to the binary machine-learning implementation.

We use four learner-class conditions. **(M1)** Conditional on its training sample, each $\hat{v}_{-k}$ is L_2 consistent and the Taylor remainder obeys

$$\sqrt{n} \sum_k \frac{|I_k|}{n} E[|R|\{\hat{v}_{-k}(D, X) - v_0(D, X)\}^2 \mid I_k^c] = o_p(1).$$

(M2) The residual-weighted projected error has an asymptotically linear representation for some mean-zero, square-integrable $\rho(W)$,

$$\sqrt{n} \sum_k \frac{|I_k|}{n} E[R\Delta_k(D, X) \mid I_k^c] = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \rho(W_i) + o_p(1). \tag{13}$$

(M3) The cross-fitted adjustment nuisances satisfy (A3), including after replacing the oracle completion by the fitted completion, and their interactions with $\hat{v}_{-k} - v_0$ are $o_p(n^{-1/2})$. **(M4)** The pass-through map is twice continuously differentiable on a neighbourhood reached with probability tending to one, with a dominated second derivative; the auxiliary uniforms are independent of the sample and one another; and the centered completion $S\{\hat{v}_{-k}\} - \psi\{\hat{v}_{-k}; v_0\}$ converges in conditional $L_2(R^2)$ to its oracle counterpart.

Condition (M2) is the substantive requirement on the imputation learner. Prediction consistency alone does not imply it because the imputation model lies outside the orthogonality guarantee. For the partially linear class, the first-order term on its left side is

$$E\Big[Rc(v_0)\left\{(\hat{\alpha} - \theta_0)D + \hat{f}(X) - g_0(X)\right\}\Big], \tag{14}$$

up to fold averaging. The treatment-coefficient error and the flexible X-direction error both enter at first order. The latter need not vanish: R is orthogonal to functions of X, but $Rc(v_0)$ generally is not.

Conjecture M1. *Under (A1) and (M1)-(M4), with fixed K and fixed B,*

$$\sqrt{n}(\hat{\theta}_B - \theta_0) = J_\theta^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} [e_i R_i + \rho(W_i) + R_i \bar{u}_i] + o_p(1), \qquad \bar{u}_i = B^{-1} \sum_{b=1}^{B} u_i^{(b)}, \tag{15}$$

and hence

$$\sqrt{n}(\hat{\theta}_B - \theta_0) \ \to_d \ N(0, V_B^{\mathrm{ML}}), \qquad V_B^{\mathrm{ML}} = J_\theta^{-2} \left\{ \mathrm{Var}(eR + \rho(W)) + \frac{\sigma_u^2}{B} \right\}. \tag{16}$$

Heuristic derivation. Conditional on the training sample for fold k, Taylor-expand Equation 12 at v_0. Proposition A1 gives the linear term in Equation 14; (M1) and (M4) control the quadratic remainder. Cross-fitting makes the empirical fluctuation of the vanishing index error $o_p(1)$, while (M2) supplies the first-order contribution from the training sample. Neyman orthogonality and (M3) remove the adjustment-nuisance terms.

At the truth, the completion error is $e_i + u_i^{(b)}$. Averaging the B completions leaves $\bar{u}_i$ with conditional variance σ_u^2/B. A joint central limit theorem then gives Equation 15 and Equation 16. This law is unconditional over the observed sample and auxiliary uniforms. Conditional on the observed sample, the uniforms represent imputation Monte Carlo error rather than repeated-sample uncertainty.

The parametric expression follows as a corollary. If $\hat{v} = v(\hat{\beta})$ is regular, then $\rho(W) = G' \operatorname{IF}_\beta(W)$ and $\operatorname{Var}(eR + \rho) = \sigma_e^2 + G'\Sigma_\beta G + 2G'C$. If each completion also redraws β, the independent draw contributes the additional $G'\Sigma_\beta G/B$ term in Equation 19. A flexible learner instead keeps its fitted index fixed across the B point-estimate completions and uses the full-pipeline bootstrap to propagate its sampling variability.

13.3.1 A proved reference learner

For a deterministic logistic-series learner, the general conjecture becomes a theorem. This learner serves as a proof reference rather than the practical default. Let

$$z_n(D, X) = \{1, D, P_n(X)'\}', \qquad \hat{v}_{-k}(D, X) = z_n(D, X)'\hat{\beta}_{-k},$$

where the dictionary P_n is fixed before observing Y and contains functions of X only. The implemented reference uses additive normalized Hermite polynomials and keeps D as a mandatory, unpenalized linear term. The population logistic projection β_n solves

$$E[z_n\{Y - \Lambda(z_n'\beta_n)\}] = 0,$$

and we write $v_n = z_n'\beta_n$,

$$A_n = E[\Lambda(v_n)\{1 - \Lambda(v_n)\}z_n z_n'], \qquad a_n = E[R\,\partial_v\psi(v_n; v_0)z_n],$$
$$\rho_n(W) = a_n' A_n^{-1} z_n\{Y - \Lambda(v_n)\}, \qquad h_n(\beta) = E[R\{\psi(z_n'\beta; v_0) - v_0\}].$$

The theorem uses five series-learner (SL) conditions. **(SL1)** The dictionary is deterministic, has dimension q_n, the eigenvalues of A_n are bounded away from zero and infinity, and the largest eigenvalue of $E[z_n z_n']$ is bounded. **(SL2)** Uniformly over unit vectors t, $E|t'z_n|^4$ is bounded, $q_n^3/n \to 0$, and $\max_i \|z_{n,i}\| = o_p\{(n/q_n)^{1/2}\}$. **(SL3)** The fold-specific logistic maximum-likelihood estimates exist uniquely with probability tending to one, so asymptotic separation is excluded. **(SL4)** The approximation is undersmoothed for the target: $\sqrt{n}\,h_n(\beta_n) \to 0$ and

$$\sqrt{n}\,E[|R|\{v_n(D, X) - v_0(D, X)\}^2] \to 0.$$

(SL5) ρ_n obeys Lindeberg's condition, $\operatorname{Var}\{\rho_n(W)\} \to \tau^2 < \infty$, and the adjustment nuisances satisfy (M3). These are standard series restrictions specialized to the residual-weighted functional. The rate and functional conditions are in the spirit of Newey (1997); bootstrap theory for semiparametric generalized additive models provides a related but different result (Härdle et al. 2004).

Theorem S1 (series expansion). *Under (A1), (M4), and (SL1)-(SL5), with fixed K and fixed B, the deterministic series learner satisfies (M2) with influence ρ_n. Consequently*

$$\sqrt{n}(\hat{\theta}_B - \theta_0) = J_\theta^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\{e_i R_i + \rho_n(W_i) + R_i\bar{u}_i\} + o_p(1), \tag{17}$$

and the limit is normal whenever the variance of the displayed triangular array converges.

Proof. Uniformly over the fixed number of folds, the fold training score and a mean-value expansion give

$$\hat{\beta}_{-k} - \beta_n = A_n^{-1}\frac{1}{|I_k^c|}\sum_{i \in I_k^c} z_{n,i}\{Y_i - \Lambda(v_{n,i})\} + r_{n,k}, \qquad \max_k |a_n' r_{n,k}| = o_p(n^{-1/2}).$$

Expanding the logistic score at β_n makes the remainder explicit. Conditions (SL1)-(SL3) give uniform convergence of the fold Hessians to A_n. The logistic third derivative is bounded, and the envelope and dimension rates in (SL2) make the quadratic score remainder $o_p(n^{-1/2})$. This step therefore does not assume the projected expansion.

A Taylor expansion of h_n at β_n has derivative a_n and quadratic remainder bounded by a constant times $E[|R|\{z_n'(\hat{\beta}_{-k} - \beta_n)\}^2]$. Conditions (SL1)-(SL2) make the fold-averaged remainder $o_p(n^{-1/2})$. In balanced K-fold cross-fitting, each observation occurs in exactly $K - 1$ training samples, and

$$\sum_k \frac{|I_k|}{n}\frac{\mathbf{1}\{i \in I_k^c\}}{|I_k^c|} = \frac{1}{n}.$$

The fold-averaged linear terms therefore equal $n^{-1}\sum_i \rho_n(W_i) + o_p(n^{-1/2})$. Condition (SL4) removes the series approximation shift at the target scale. The generated-outcome Taylor remainder is also negligible by (SL4) and (M4). Finally, (M3) removes the adjustment-nuisance terms, while the completion split in Equation 9 supplies $e_i R_i + R_i \bar{u}_i$. Lindeberg's theorem and (SL5) give the stated normal limit. □

The target-undersmoothing clause in (SL4) is essential. If only its quadratic-approximation clause holds, the same argument centers the estimator at the series-projection target

$$\theta_n = \theta_0 + J_\theta^{-1} h_n(\beta_n),$$

not at θ_0. The fixed degree-three implementation below measures this population shift directly. A growing dictionary can satisfy the undersmoothing condition; a convenient fixed dictionary should not be treated as automatically eliminating approximation bias.

13.4 Bootstrap propagation for the learner class

Because the fixed-B estimator is randomized, its bootstrap statement differs slightly from that for a deterministic asymptotically linear estimator. Let $\hat{\theta}_B^*$ rerun the entire procedure on an Efron resample, including fresh surrogate uniforms, and let

$$\mu_n^* = E^*(\hat{\theta}_B^* \mid \mathcal{D}_n)$$

average over resampling, any redrawn fold partition, and the fresh surrogate randomness. We add four machine-learning bootstrap (MB) conditions. **(MB1)** The projected learner expansion in (M2) is bootstrap stable: after centering, its resampled version has influence $n^{-1/2}\sum_i (N_i - 1)\rho(W_i) + o_{P^*}(1)$ in outer probability. **(MB2)** The adjustment-nuisance rates in (M3) hold conditionally under resampling. **(MB3)** The integrated map ψ has the dominated differentiability in (M4), and the centered randomized completion is conditionally $L_2(R^2)$ continuous in the fitted index. **(MB4)** Fold sizes are asymptotically balanced and the learner conditions hold uniformly over either a fixed partition independent of the observations or a fresh regular partition in every resample.

The conditions clarify how two existing bootstrap results apply here. Tang and Westling (2024) give general consistency conditions for estimators with machine-learned nuisances, and Lin and Han (2026) establish exchangeably weighted bootstrap validity for DML under its usual conditions. SUDO additionally requires the generated-outcome conditions (MB1) and (MB3). A raw inverse-transform draw need not be differentiable for every fixed uniform at a truncation boundary. Its integrated score ψ must be differentiable, and conditional L_2 continuity controls the centered-draw remainder.

Conjecture M2. *Under (A1), (M1)-(M4), and (MB1)-(MB4), with fixed K and fixed B,*

$$\sup_t \left| P^*\{\sqrt{n}(\hat{\theta}_B^* - \mu_n^*) \le t\} - \Phi\{t/\sqrt{V_B^{\mathrm{ML}}}\} \right| \to_p 0, \qquad n \ \mathrm{Var}^*(\hat{\theta}_B^*) \to_p V_B^{\mathrm{ML}}. \tag{18}$$

Heuristic derivation. Apply the exchangeably weighted expansion to $eR + \rho(W)$ using (MB1)-(MB2). Conditional on each resampled observation, the fresh centered completion terms are independent across observations and draws, with limiting contribution σ_u^2/B; (MB3) permits replacing the fitted index in that term by its limit. The conditional Lindeberg central limit theorem and (MB4) give the same law for fixed or redrawn regular folds. Conditional second-moment convergence gives the variance statement.

Theorem S2 (series full-pipeline bootstrap). *Suppose the conditions of Theorem S1 hold uniformly on Efron resamples, the empirical Hessian and score moments in (SL1)-(SL3) converge conditionally, and (MB2)-(MB4) hold. Also suppose the empirical second moment of $\mathrm{e}_i R_i + \rho_n(W_i)$ converges to its population counterpart and its maximum absolute value is $o_p(\sqrt{n})$. Then the deterministic series learner satisfies (MB1). Consequently Equation 18 holds for this learner, for either fixed or independently redrawn regular folds.*

Proof. On a resample with multiplicities N_i, expand each fold training score at its original-sample solution. Conditional convergence of the resampled Hessian and the same bounded logistic derivative used in Theorem S1 give

$$\sum_k \frac{|I_k|}{n}\{h_n(\hat{\beta}_{-k}^*) - h_n(\hat{\beta}_{-k})\} = \frac{1}{n}\sum_{i=1}^n (N_i - 1)\rho_n(W_i) + o_{P^*}(n^{-1/2})$$

in outer probability. The balanced-fold identity from Theorem S1 applies conditionally for a fixed partition. For an independently redrawn regular partition, (MB4) makes the difference between its inclusion weights and $1/n$ negligible at the same scale. This proves (MB1) rather than assuming it. The exchangeably weighted central limit theorem applies to $eR + \rho_n$; (MB2) removes the adjustment-nuisance remainder, and (MB3) permits replacement of the fitted index in the centered completion term by its limit. Fresh completion terms contribute σ_u^2/B conditionally. Conditional second-moment convergence yields bootstrap-variance consistency. □

The centering in Equation 18 affects the inferential conclusion. At fixed B, $\hat{\theta}_B - \mu_n^*$ contains $O_p(n^{-1/2})$ surrogate Monte Carlo noise. Consequently the usual claim with $\hat{\theta}_B^* - \hat{\theta}_B$ does not follow. The bootstrap standard deviation is still consistent and therefore supports the normal interval $\hat{\theta}_B \pm z_{1-\alpha/2}\hat{s}_{\text{boot}}$. A centered bootstrap distribution may instead estimate μ_n^* by the mean of the outer replicates. Raw percentile and conventionally centered studentized intervals require either $B \to \infty$ fast enough to remove this centering gap or an additional randomized-estimator argument. We report them as empirical comparisons, not consequences of Conjecture M2.

Neither (M2) nor its bootstrap analogue (MB1) has been proved separately for the implemented GAM, MARS, or neural-backfit algorithms. The conjectures therefore identify the learner-specific obligations; numerical performance does not replace those proofs. Redrawing folds is also unnecessary when (MB4) holds, although it remains the implementation default because it propagates partition variability. At its Monte Carlo resolution, the paired check in Section 15 finds fixed and redrawn folds indistinguishable.

13.5 Moment properties

Proposition A2. *Under (A1)-(A2), $E[\varphi] = 0$ at the truth and the moment is Neyman-orthogonal in the adjustment nuisances, $\partial_\ell E[\varphi] = \partial_m E[\varphi] = 0$. It is not orthogonal in β: $\partial_\beta E[\varphi] = G$ with blocks*

$$G_{\beta_\eta} = E[c(\eta_0)\,\partial_{\beta_\eta}\eta\, R], \qquad G_{\kappa_k} = E[f(\kappa_k^0 - \eta_0)\,(\mu_{k+1} - \mu_k)\, R].$$

Proof. At the truth $S - \ell_0 - \theta_0 R = \xi$, and averaging over $Y \mid D, X$ makes $\xi \mid D, X \sim F$, so $E[\varphi] = E[\xi R] = 0$. The mean-zero statement holds only after that averaging; given (W, β) the draw has mean $e \neq 0$. The nuisance derivatives vanish by the tower property. The β derivative is Equation 11 composed with $\partial_\beta \eta$. □

Because R annihilates any function of X, profiling ℓ out of the moment leaves G unchanged. This result covers both implementations: refitting $\hat{\ell}$ for every completion, or fitting it once from a plug-in completion and holding it fixed, as the binary estimator does. The appendix section on nuisances and draws explains this distinction. Proposition A2 formalizes the claim in the body that the imputation model lies outside the orthogonality guarantee.

13.6 Asymptotic distribution at fixed B

Conjecture A1. *Under (A1)-(A5), with $\bar{\theta} = B^{-1}\sum_b \hat{\theta}^{(b)}$,*

$$\sqrt{n}(\bar{\theta} - \theta_0) \to_d N(0, V_B), \qquad V_B = J_\theta^{-2}\Big[\sigma_e^2 + G'\Sigma_\beta G + 2G'C + \frac{G'\Sigma_\beta G + \sigma_u^2}{B}\Big], \tag{19}$$

where $C = \mathrm{Cov}(eR, \mathrm{IF}_\beta)$.

Heuristic derivation. The Z-estimator expansion gives $\sqrt{n}(\hat{\theta} - \theta_0) = J_\theta^{-1}\sqrt{n}M_n(\theta_0, \cdot) + o_p(1)$, the sign following from $\partial_\theta E[\varphi] = -J_\theta$. Expanding around (β_0, ℓ_0, m_0), the adjustment-nuisance terms are $o_p(1)$ by Proposition A2 plus cross-fitting plus the (A3) product rate, as in Chernozhukov et al. (2018). The β term is $G'\sqrt{n}(\beta^{(b)} - \beta_0) + o_p(1)$, and $\beta^{(b)} - \beta_0 = (\hat{\beta} - \beta_0) + n^{-1/2}Z_b$ by (A5). Applying Equation 9 separates the data-measurable score $e_i R_i$ from the draw noise; averaging over b leaves parameter-draw and surrogate-draw terms with conditional variances Σ_β/B and σ_u^2/B.

Two features of Equation 19 are important for interpretation. First, it is not the single-completion result plus an $O(1/B)$ correction because the leading term itself depends on B. A single completion carries the full sandwich $\sigma_e^2 + \sigma_u^2$ *and* twice the parameter contribution, once from $\hat{\beta}$'s own sampling error and once from the single draw around it, which is exactly what Rubin's $(1 + 1/B)$ factor exists to undo. As $B \to \infty$ the draw noise averages away and the leading term is σ_e^2, not $\sigma_e^2 + \sigma_u^2$. Second, the cross term $2G'C$ is not sign-restricted because IF_β and the moment are built from the same data. Consequently, V_B is not the DML sandwich plus a nonnegative imputation penalty. Pooling removes the draw noise from the sandwich and adds $G'\Sigma_\beta G + 2G'C$.

13.7 The pooled variance

Rubin's total is $T = \bar{W} + (1 + 1/B)B_*$, but its two components behave differently. The within-draw term $\bar{W}$ averages a sandwich estimate over n observations, so $n\bar{W} \to_p J_\theta^{-2}(\sigma_e^2 + \sigma_u^2)$. The between-draw term B_* is the sample variance of B draws. Writing $V_{\text{draw}} = G'\Sigma_\beta G + \sigma_u^2$ for the per-draw variance, Equation 19 gives

$$nB_* \to_d J_\theta^{-2}\, V_{\text{draw}}\, \frac{\chi^2_{B-1}}{B-1} \qquad (n \to \infty,\ B \text{ fixed}), \tag{20}$$

which is *not* a probability limit. With B fixed, the between-draw term retains sample-variance randomness regardless of the sample size. It is conditionally unbiased for $J_\theta^{-2} V_{\text{draw}}$, allowing the comparison below to be stated in expectation.

Write $E_\Delta[\cdot] = E[\cdot \mid W_1, \dots, W_n]$ for the expectation over the imputation draws with the data held fixed, so that $E_\Delta[T]$ is still a random variable through the data.

Conjecture A2. *Under (A1)-(A5), for fixed B,*

$$n\, E_\Delta[T] - V_B \;\to_p\; 2J_\theta^{-2}(\sigma_u^2 - G'C),$$

with nT itself converging in distribution, not in probability, by Equation 20. Taking expectations over the data as well gives the same limit for the unconditional $n\,E[T] - V_B$, which is the form the Monte Carlo check below estimates, since each replication redraws both the sample and the imputations. As $B \to \infty$ the randomness in B_* averages away and the statement strengthens to a probability limit for T itself,

$$n\,T - V_\infty \;\to_p\; 2J_\theta^{-2}(\sigma_u^2 - G'C), \qquad V_\infty = J_\theta^{-2}(\sigma_e^2 + G'\Sigma_\beta G + 2G'C).$$

Either way the $1/B$ terms cancel, so the discrepancy is free of B, and Rubin's variance is asymptotically exact if and only if $G'C = \sigma_u^2$, conservative when $G'C < \sigma_u^2$, and anti-conservative when $G'C > \sigma_u^2$.

Heuristic derivation. Substitute the probability limit of $n\bar{W}$ and the conditional mean of Equation 20 into Rubin's total, then subtract Equation 19. The $1/B$ terms cancel. The $B \to \infty$ statement additionally uses concentration of the between-draw sample variance.

The joint law determines the appropriate reference distribution. Define

$$V_{\text{data}} = \sigma_e^2 + G'\Sigma_\beta G + 2G'C, \quad W_0 = J_\theta^{-2}(\sigma_e^2 + \sigma_u^2), \quad Q_0 = J_\theta^{-2}(1 + 1/B)V_{\text{draw}},$$

and let $\nu = B - 1$.

Conjectured fixed-B reference law. *Under (A1)-(A5),*

$$\left\{\sqrt{n}(\bar{\theta} - \theta_0),\, nT\right\} \;\to_d\; \left\{\sqrt{V_B}\, Z,\, W_0 + Q_0 \frac{X}{\nu}\right\}, \qquad Z \sim N(0,1), \quad X \sim \chi^2_\nu, \quad Z \perp X. \tag{21}$$

Consequently

$$\frac{\bar{\theta} - \theta_0}{\sqrt{T}} \;\to_d\; \frac{\sqrt{V_B}\, Z}{\sqrt{W_0 + Q_0 X/\nu}}. \tag{22}$$

Heuristic derivation. The joint central limit theorem posited in Conjecture A1 writes the bth completed-data limit as $J_\theta^{-1}(A + Q_b)$, where $A \sim N(0, V_{\text{data}})$ is common to the draws and the $Q_b \overset{\text{iid}}{\sim} N(0, V_{\text{draw}})$ are independent of A. Their sample mean enters $\bar{\theta}$ and their sample variance enters B_*. Normal sample means and sample variances are independent, while $n\bar{W} \to_p W_0$, which gives Equation 21.

The exact limiting critical value is therefore a quantile of a normal-over-shifted-chi-square mixture, not generally a Student t. It is obtained by solving

$$E_X\left[2\Phi\left\{q_B \sqrt{\frac{W_0 + Q_0 X/\nu}{V_B}}\right\} - 1\right] = 1 - \alpha. \tag{23}$$

A convenient approximation matches the first two moments of the random denominator. With

$$\mu_T = W_0 + Q_0, \qquad \nu_S = \nu(\mu_T/Q_0)^2,$$

the variance-calibrated Satterthwaite critical value is $q_S = \sqrt{V_B/\mu_T}\, t_{1-\alpha/2,\nu_S}$. Barnard-Rubin (BR) instead uses the observed $r = (Q_0 X/\nu)/W_0$ and $\nu_{\text{BR}} = \nu(1 + 1/r)^2$ in the limit. It is not exactly Equation 22, but the comparison in Section 15 shows that it is close.

The discrepancy need not have a fixed sign, and the calculation does not yield exactness. On a deliberately congenial testbed that isolates pooling from every other source of error (a correctly specified logistic imputation model, continuous treatment so both adjustment nuisances are exactly linear), all six quantities are available in closed form:

θ_0	σ_e^2	σ_u^2	$G'\Sigma_\beta G$	$G'C$	$\sigma_u^2 - G'C$	T/V at $B = 20$
1	1.269	2.021	4.077	1.954	+0.067	1.014
3	0.580	2.710	41.414	3.094	−0.384	0.985

The sign of the discrepancy flips with signal strength: on average T overstates the sampling variance by about 1.4% at $\theta_0 = 1$ and understates it by about 1.5% at $\theta_0 = 3$. The T/V column compares $n\,E[T]$ with V_B, which is the quantity Conjecture A2 governs at fixed B; individual replications scatter around it according to Equation 20. Both discrepancies are asymptotic rather than finite-sample effects; they persist at $n = 20{,}000$ and, being free of B, do not shrink as B grows. The mechanism is visible in the table. As the signal grows the observed Y carries less information beyond the index, so σ_e^2 falls while σ_u^2 and $G'C$ rise, and $G'C$ overtakes σ_u^2. This is a more precise account of why inference is hardest at strong effect sizes than the informal argument in the body, and it corrects a natural but mistaken reading: $c \to 1$ is the regime in which ψ is *closest* to the identity, so the linearization is at its most accurate there, and cannot be what degrades.

The practical effect is small: a 1.5% variance error corresponds to roughly 0.7% in a standard error, well within the Monte Carlo error of the coverage figures in Section 8. The theoretical conclusion is nevertheless important: proper draws plus Rubin's rules are not exactly variance-calibrated here. Section 15 reports the term-by-term Monte Carlo check of Equation 19.

Implication for improper draws. If $\beta^{(b)} \equiv \hat{\beta}$, the between-draw term loses the parameter contribution entirely, so B_* is conditionally unbiased for $J_\theta^{-2}\sigma_u^2$ alone and $n\,E[T]$ falls short of V_B. The deficit is dominated by the imputation channel $G'\Sigma_\beta G$ whenever $G \neq 0$ and the signal is not weak, though it is not exactly that quantity, since the cross term and the draw noise also enter. This is the 0.93 versus 0.97 coverage gap in Table 1.

13.8 Bias under a misspecified imputation model

Proposition A3. *Consider a local sequence of binary-treatment imputation models whose index error equals δ_D in the treated arm and 0 in the control arm, with $\delta_D \to 0$. The corresponding population SUDO target $\theta(\delta_D)$ satisfies*

$$\theta(\delta_D) - \theta_0 = \bar{c}_1 \delta_D + o(\delta_D), \qquad \bar{c}_1 = \frac{E[w\,c(\theta_0 + g_0)]}{E[w]}, \quad w = \mathrm{Var}(D \mid X).$$

Proof. Differentiate the generated-outcome mean $E[S \mid D, X]$ with respect to the local index perturbation at zero. Proposition A1 gives derivative $c(\eta_0)$. Applying $J_\theta^{-1}E[\,\cdot\,R]$ and conditioning on X with binary D, the moment weights the treated arm by w and the control arm by $-w$; the control-arm error is zero by assumption, so only $c(\theta_0 + g_0)$ survives, and $J_\theta = E[w]$ supplies the denominator. □

Both restrictions matter. The displayed $\bar{c}_1$ evaluates c on the treated arm only, which is correct precisely because the control-arm error was set to zero; a general perturbation weights both arms as $m\,c(\theta_0 + g_0) + (1 - m)\,c(g_0)$. For continuous D with index error $\delta_D D$ the multiplier is instead $\bar{c}_1 = E[c(\eta_0)DR]/E[R^2]$, which is a different expression. With a simultaneous covariate-direction perturbation, the first-order leak is Equation 7, nonzero because c depends on D through η and so is not annihilated by projecting on R. Table 3 is the empirical counterpart for the treatment direction, recovering the predicted slope $\bar{c}_1$.

For a finite perturbation, define the arm-specific path derivative and its average

$$c_d(t; h) = \partial_v \psi(v; \eta_d)\big|_{v = \eta_d + th}, \qquad \bar{c}_d(h) = \int_0^1 c_d(t; h)\,dt,$$

where $\psi(v; \eta_d)$ uses fitted index v while the observed category is still generated at the true index η_d. At $h = 0$, $c_d(0; 0) = c(\eta_d)$ from Proposition A1. Write $\|a\|_w = \{E[wa^2]\}^{1/2}$ and $J_\theta = E[w]$.

Proposition A4 (covariate-leakage bound). *Suppose D is binary and the limiting imputation-index error is the same $h(X)$ in both arms. Then the population SUDO target has the exact covariate-direction bias*

$$L_X(h) = J_\theta^{-1} E[w\,h\{\bar{c}_1(h) - \bar{c}_0(h)\}]\,, \tag{24}$$

and hence

$$|L_X(h)| \le J_\theta^{-1} \|h\|_w \,\|\bar{c}_1(h) - \bar{c}_0(h)\|_w. \tag{25}$$

Writing the first-order term as

$$L_X^{(1)}(h) = J_\theta^{-1} E[w\,h\{c(\eta_1) - c(\eta_0)\}],$$

the sharper local bound is

$$|L_X(h) - L_X^{(1)}(h)| \leq J_\theta^{-1} \|h\|_w \, \|\{\bar{c}_1(h) - c(\eta_1)\} - \{\bar{c}_0(h) - c(\eta_0)\}\|_w \,. \tag{26}$$

Proof. The fundamental theorem of calculus gives $\psi(\eta_d + h; \eta_d) - \eta_d = h\bar{c}_d(h)$. Conditioning $E[R\Delta_D]$ on X gives $w(\Delta_1 - \Delta_0)$, proving Equation 24. The two bounds are Cauchy-Schwarz, first on the full path-factor difference and then only on its deviation from the derivative at the truth. □

The bound is computable for a specified sensitivity perturbation h and becomes a conservative sensitivity radius when $\|h\|_w$ is bounded. Under $\epsilon \leq m(X) \leq 1-\epsilon$, $J_\theta \geq \epsilon(1-\epsilon)$ makes the overlap dependence explicit. The local remainder is second order when the path derivative is locally Lipschitz. Nothing forces the first-order term or its bound to be small relative to the treatment-direction contribution.

14 Appendix: simulation and application details

Each design draws a latent utility $U = \theta_0 D + g_0(X) + \varepsilon$ and converts it to an observed category. Most designs use logistic ε; the exceptions are specified below. The designs vary the covariate structure, cutpoints, and fitted models; covariates remain standard normal throughout. The `R/` and `R/results/` directories contain the full data-generating code and committed per-cell results.

14.1 Data-generating processes

All designs instantiate Equation 8 with one of two covariate structures:

- **S1**, one covariate: $p = 1$, $\pi(X) = 0.8X$, $g_0(X) = X$. Every model in play is correctly specified, so only the pooling is under test.
- **S5**, five covariates: $p = 5$, $\pi(X) = X_4 + \cos X_5$, $g_0(X) = X_1^2 + \sin X_2 + 0.5X_3$. The outcome structure is nonlinear in X, and treatment depends on covariates absent from g_0.

Table 8: Data-generating processes. κ lists the interior cutpoints, so (0) is binary and a pair gives $J = 3$.

Design	Structure	F	κ	θ_0	n	Replications
Inference ladder (Table 1)	S1	logistic	(0)	1.5	5000	500
Ordinal parametric (Table 2, top)	S1	logistic	$(-1, 1)$	1.0	3000	500
Binary ML	S5	logistic	(0)	0.5, 3	1000, 2000	200
Ordinal ML (Table 2, bottom)	S5	logistic	$(0, 2)$	1.0	1000, 2000	200
Link misspecification	S1	Gumbel	(0) or $(0, 2)$	1.0	5000	200
Sensitivity (Table 3)	S5	logistic	(0)	1.5, 3	2000	200
Black-box imputation	S5	logistic	(0)	1.5, 3	2000	100
PL learner comparison (Table 4)	S5	logistic	(0)	1.5, 3	2000	50
Bayesian imputation (repository only)	S5′	normal	(0)	1.5, 3	2000	200
DGP variation, continuous D	S5, $D = 0.7\pi(X) + N(0,1)$	logistic	(0) or $(0, 2)$	1, 2.5	2000	200

Design	Structure	F	κ	θ_0	n	Replications
DGP variation, interaction	S5 plus X_1X_2 in g_0	logistic	(0)	1, 2.5	2000	200
DGP variation, dimension	S5 with $p = 10$	logistic	(0)	1, 2.5	2000	200
DGP variation, overlap	S5, treatment-assignment index $\pi(X)$ scaled $2.5\times$	logistic	(0)	1, 2.5	2000	200

S5′ is the probit analogue of S5 used for the Bayesian imputation model, whose binary machinery is probit: the systematic part is divided by $\mathrm{sd}(\mathrm{logistic}) = \pi/\sqrt{3}$ before normal errors are added, which matches the event rates of S5 to within 0.006 and puts θ_0 on the probit scale at $\theta_0/1.8138$. The θ_0 column reports the logit-scale values for comparability. That design supports no table here; it is reported so the claim in the machine-learning section is checkable.

Models fitted. The ladder uses $\mathrm{glm}(Y \sim D + X)$ as the imputation model with $B \in \{10, 25, 50\}$ draws (the table shows $B = 25$). The ordinal parametric design uses `ordinal::clm`, whose covariance covers the thresholds, with $B = 25$. The ordinal ML design uses `clm` on D and natural-spline bases $\mathrm{ns}(X_k, 4)$ with `mgcv::gam` partialling nuisances and $B = 25$.

Link misspecification. This design retains the one-covariate structure but replaces logistic ε with an extreme-value error: Gumbel-max for the binary case, matching `glm` with a cloglog link, and Gumbel-min for the ordinal case, matching `ordinal::clm` with a cloglog link. The analyst fits either logit (misspecified) or cloglog (correct). Holding the simple covariate structure fixed isolates error in the link.

Sensitivity ladder (Table 3). The ladder starts from the oracle index and oracle partialling nuisances, then adds a known perturbation of size δ to one input: the outcome nuisance only (DML-1), both nuisances (DML-2), the index in the covariate direction (SUDO-X), or the index in the treatment direction (SUDO-D). Bias is traced against a common latent-scale root-mean-square error (RMSE) so the four are comparable. The fitted SUDO-D slope reproduces $\bar{c}_1$ (0.669 at $\theta_0 = 1.5$, 0.861 at $\theta_0 = 3$) to within 0.001, which is the independent check on Equation 5.

Partially-linear imputation models. The original black-box result uses $\widehat{V} = \hat{\alpha} D + \widehat{f}(X)$ with $\widehat{f}$ a single-hidden-layer neural net (`nnet`, `size = 4`, `decay = 0.3`, five backfitting iterations, selected on target-level Monte Carlo bias) and 100 full-pipeline bootstrap resamples. The common adaptive-learner comparison uses 50 resamples for each arm. Its GAM has an unpenalized linear treatment term and smooths of the five covariates. The MARS fit builds degree-two bases from X only (`nk = 21`, penalty 3), then jointly estimates those basis coefficients and a mandatory linear treatment coefficient by logistic regression. This construction is needed because `earth::linpreds` alone still permits the named predictor in interaction terms. The theorem-reference series is evaluated separately. It uses additive normalized Hermite polynomials through degree three, a dictionary fixed before observing Y, and a separate mandatory linear treatment term.

Wine application. Full model `ordinal::clm` on D and either the raw covariates (linear) or $\mathrm{ns}(X_k, 3)$ (spline); partialling nuisances $E[S \mid X]$ and $E[D \mid X]$ by `mgcv::gam` (Gaussian for the continuous D); five-fold cross-fitting; $B = 50$ proper draws; Rubin pooling. Volatile acidity is standardized.

14.2 Nuisances, cross-fitting, and draws

Partialling nuisances $\widehat{\ell}(X) = E[S \mid X]$ and $\widehat{m}(X) = E[D \mid X]$ are `mgcv::gam` throughout, fit with five-fold cross-fitting. Proper draws sample the full-model parameters from $N(\widehat{\beta}, \widehat{\mathrm{Cov}}(\widehat{\beta}))$ (thresholds included for `clm`) before each completion. Diagnostic draws use a random number generator (RNG) stream independent of the estimation draws.

One algorithmic detail differs by outcome type. Because the surrogate S changes with every draw, its nuisance $\widehat{\ell}$ can also be refit each time. Binary experiments found this refitting negligible, so the binary estimator fits $\widehat{\ell}$ once on a plug-in completion and reuses it. The ordinal nuisance-isolation ladder gives a different result: holding $\widehat{\ell}$ fixed nearly doubles the within-draw component and accounts for the stage-5 conservatism. The corrected ordinal estimator therefore cross-fits $\widehat{\ell}$ separately for every completion. The full validation with $B = 25$ and 200 replications attains 0.960 coverage.

Refitting costs roughly B times as much for the outcome nuisance, so the less expensive binary behavior is retained where it has been validated.

14.3 Monte Carlo precision

Coverage is a binomial proportion, so its Monte Carlo standard error at a true 0.95 is about $\sqrt{0.95 \cdot 0.05/R}$: 0.010 at $R = 500$ replications, 0.015 at $R = 200$, and 0.022 at $R = 100$. Per-cell bias and coverage Monte Carlo standard errors are in the committed result files.

15 Supplement: numerical verification of the theory

Every closed-form expression in Section 13 is checked through an independent numerical route. The corresponding scripts are in the project repository and assert their own pass conditions.

15.1 Pass-through derivatives (Proposition A1)

`R/theory_ordinal_passthrough.R` computes ψ by quadrature and compares the analytic index and threshold derivatives of Equation 11 against central differences, for $J = 2, 3, 4$ under logit, probit, and both Gumbel conventions. Maximum discrepancy across all configurations is 3×10^{-10} for the index block and 4×10^{-10} for the threshold block; the identity $c = 1 - \sum_k \partial_{\kappa_k} \psi$ holds to 10^{-15}. A Monte Carlo check confirms `complete_surrogate()` samples the map it should ($|z| < 4$ at 4×10^5 draws). The analytic map therefore matches the implemented estimator.

At $e = 0$ under a logit, the threshold-placement effect discussed in Section 13 is:

κ	(0) binary	$(-0.5, 0.8)$	$(-1, 0.2, 1.3)$	(∓ 0.5)	(∓ 1.5)	(∓ 3)	(∓ 6)
$c(0)$	0.307	0.157	0.090	0.175	0.223	0.636	0.966

The final four entries all use $J = 3$. Widening the three-category partition moves c from well below the binary value to nearly 1.

15.2 Deterministic-series expansion (Theorems S1 and S2)

`R/theory_pl_series.R` checks the proof-specific quantities without using the fitted SUDO estimate as its own benchmark. It computes the population logistic-series projection and its influence function by product Gauss-Hermite quadrature, then compares them with independently fitted five-fold estimates. The implemented dictionary uses normalized additive Hermite polynomials through degree three and a separate mandatory linear treatment term.

The population projection shifts the SUDO target by -0.00082 at $\theta_0 = 1.5$ and -0.00167 at $\theta_0 = 3$. These small values are measured approximation errors for this design, not a substitute for (SL4). At $n = 4000$, the empirical standard deviation divided by the influence-function prediction is 0.989 and 0.985 at the two signals. The root-mean-square expansion remainder divided by the predicted standard deviation falls from 0.439 to 0.198 at the moderate signal and from 0.623 to 0.289 at the strong signal as n increases from 1000 to 4000. The strong-signal remainder is visibly slower to disappear, consistent with the near-separation concern isolated in (SL3).

For 30 datasets at $n = 2000$ and 50 Efron resamples per dataset, the bootstrap standard-deviation ratio to the weighted influence expansion is 1.015 with fixed folds and 1.014 with redrawn folds at $\theta_0 = 1.5$; the corresponding ratios are 1.076 and 1.077 at $\theta_0 = 3$. All 3000 paired bootstrap resamples are complete. These checks validate the algebra and finite-sample behavior of the theorem reference. Theorems S1 and S2 provide the proof; the simulation does not.

15.3 Machine-learning expansion (Conjecture M1)

`R/theory_ml_expansion.R` compares the exact conditional generated-outcome shift $\psi(\hat{v}; v_0) - v_0$ with its linear approximation $c(v_0)(\hat{v} - v_0)$ for GAM, MARS, and neural backfitting. It uses the S5 design, $\theta_0 \in \{1.5, 3\}$, $n \in \{1000, 2000\}$, and 50 replications per cell. The analytic derivative agrees with central differences to 7.2×10^{-11}. Across all 12 learner cells, the mean absolute pointwise Taylor remainder divided by mean squared index error ranges from 0.016 to 0.027. At $n = 2000$, the absolute residual-projected remainder ranges from 0.0015 to 0.0107, against a first-order scale of 0.063 to 0.138. The generated-outcome part of (M1) is therefore second order in the fitted-index error in every implemented design. This check does not establish the learner-specific asymptotic representation in (M2).

15.4 Machine-learning bootstrap (Conjecture M2)

`R/theory_ml_bootstrap.R` compares fixed and redrawn folds on the same 30 generated datasets for each learner and signal, with 15 outer resamples and 15 completions per resample. The fixed/redrawn mean-SE ratio ranges from

0.97 to 1.08, and every paired bias difference is smaller than its prespecified Monte Carlo tolerance. At this resolution, redrawing folds is not empirically necessary in this design.

The uncertainty hierarchy depends on signal strength but is consistent across learners. At $\theta_0 = 1.5$, the within-fold SE is 0.90 to 0.93 of the full-pipeline SE; at $\theta_0 = 3$ it is only 0.75 to 0.77. Holding the fitted index fixed reduces the SE further, to 0.75 to 0.92 of the within-fold value. The within-fold method therefore does not reproduce the improper method. It captures some learner uncertainty but misses the data, nuisance, and fold layers propagated by the full pipeline. Normal coverage is 0.93 to 0.97 across the six cells. Percentile coverage is 0.87 to 0.97 and conventionally centered studentized coverage is 0.80 to 0.93, which supports the fixed-B centering qualification rather than either alternative as a default. With 30 replications, coverage Monte Carlo standard errors are about 0.03 to 0.07. These checks support the implications of (MB4) and the variance ordering, but do not establish learner-specific condition (MB1).

15.5 The variance decomposition (Conjectures A1 and A2)

`R/theory_variance_terms.R` evaluates J_θ, σ_e^2, σ_u^2, G, Σ_β and C in closed form on the congenial testbed by product-grid quadrature, then compares Equation 19 against 4000 replications at $n = 20{,}000$ for $B \in \{1, 2, 5, 20\}$ and $\theta_0 \in \{1, 3\}$. All eight cells agree within 1.2 Monte Carlo standard errors. Comparing across B is what identifies the individual terms; a single pooled variance ratio cannot, because its Monte Carlo error is the same size as the 1 to 2 percent effect being measured, which is why the earlier ratio-based reading of these simulations was inconclusive rather than confirmatory.

The fixed-B spread of Equation 20 is confirmed directly: at $\theta_0 = 1$, $B = 20$, $n = 20{,}000$, the observed standard deviation of nT across replications is 2.13 against a mean of 9.73, while Equation 20 predicts $(1+1/B)V_{\text{draw}}\sqrt{2/(B-1)} = 2.08$. The residual spread in the reported variance is therefore attributable to the between-draw sample variance rather than estimation error, and it does not shrink with n.

`R/theory_fixed_b_inference.R` evaluates Equation 22 at $B \in \{5, 10, 25, 50, 100\}$ and then checks it with 1,000 independent samples at $n = 5{,}000$. Prediction and empirical coverage differ by at most 0.012 across all 40 comparisons in Table 10. Normal inference is inadequate at small B and strong signal; Barnard-Rubin is close to the exact mixture, and calibrated Satterthwaite coverage is within 0.006 of 0.95 in every cell.

Table 10: Fixed-imputation 95% coverage, theory prediction / finite-sample estimate; $n = 5000$, 1000 replications.

θ_0	B	Normal	Barnard-Rubin	Satterthwaite	Mixture
1	5	0.923 / 0.924	0.945 / 0.950	0.956 / 0.954	0.950 / 0.950
1	10	0.939 / 0.941	0.950 / 0.952	0.951 / 0.952	0.950 / 0.951
1	25	0.947 / 0.944	0.951 / 0.951	0.950 / 0.951	0.950 / 0.950
1	50	0.949 / 0.947	0.952 / 0.948	0.950 / 0.948	0.950 / 0.947
1	100	0.950 / 0.953	0.952 / 0.954	0.950 / 0.953	0.950 / 0.953
3	5	0.888 / 0.880	0.942 / 0.933	0.954 / 0.952	0.950 / 0.949
3	10	0.921 / 0.913	0.947 / 0.948	0.951 / 0.952	0.950 / 0.952
3	25	0.938 / 0.950	0.948 / 0.955	0.950 / 0.956	0.950 / 0.956
3	50	0.943 / 0.953	0.948 / 0.957	0.950 / 0.959	0.950 / 0.959
3	100	0.946 / 0.952	0.948 / 0.956	0.950 / 0.959	0.950 / 0.959

The practical default remains Barnard-Rubin with at least 25 imputations. It requires only the usual pooled quantities and is close to the exact mixture, whereas the calibrated alternatives require consistent estimates of V_B, W_0, and Q_0. Increasing B reduces the random chi-square component but does not remove Conjecture A2's systematic Rubin discrepancy. For a flexible imputation learner, the full-pipeline bootstrap normal interval remains the corresponding default; the percentile and conventional studentized comparisons in Table 5 do not improve it.

15.6 Covariate-direction leakage (Proposition A4)

`R/theory_covariate_leakage.R` evaluates the exact path identity and both bounds with two million population draws from the stage-7 interaction design. It omits $\lambda X_1 X_2$ for $\lambda \in \{\mp 0.5, \mp 1, \mp 1.5\}$ at $\theta_0 \in \{1, 2.5\}$. The global and local bounds cover the exact covariate leak in all 12 cells. The local bound is 2.82 to 7.79 times the absolute leak, while the absolute Taylor remainder divided by λ^2 is 0.0027 to 0.0091. It is therefore conservative but has the correct second-order scale in this symmetric interaction design.

The calculation also corrects the earlier interpretation of stage 7. Omitting the interaction shifts the pseudo-true logistic treatment coefficient even though D is constrained to enter linearly. At $\lambda = 1$, the population decomposition is:

θ_0	Covariate leak	Induced treatment component	Total prediction	Stage-7 bias
1.0	-0.015	-0.021	-0.036	-0.050
2.5	-0.013	-0.068	-0.081	-0.147

The population calculation is deliberately simpler than the fitted additive GAM and understates the observed magnitude, especially at strong signal. Its differences from stage 7, 0.014 and 0.066, are inside the prespecified Monte Carlo tolerances of 0.030 and 0.075. More importantly, the decomposition establishes that the interaction experiment is not a pure covariate-leakage arm: the induced treatment-direction channel is at least as large.

Both closed forms behind these quantities rest on one identity, $p(1-p)(\mu_+ - \mu_-) = H(p)$ for the logistic law, which yields both $c(e) = 1 - H(F(e))$ and $C = \Sigma_\beta(A - G)$ with $A = E[zR]$. It is verified pointwise by quadrature to 8×10^{-16}.

The ordinal counterpart is checked by `R/theory_ordinal_variance_terms.R` on a correctly specified three-category cumulative-logit design with continuous treatment and exactly linear partialling nuisances. The calculation includes both threshold derivatives in G and matches the fixed-B sampling variance and expected Rubin variance within 2.3 Monte Carlo standard errors in every cell at $n = 8{,}000$, 500 replications, $B \in \{1, 2, 5, 20\}$, and $\theta_0 \in \{1, 2.5\}$. At $B = 20$, the predicted T/V is 1.028 and 0.980, respectively. The ordinal threshold machinery therefore changes the same small Rubin discrepancy identified in the binary testbed but does not explain the much larger stage-5 gap.

`R/theory_ordinal_nuisance_ladder.R` then changes one partialling component at a time on the exact stage-5 DGP, using common samples, folds, imputation fits, parameter draws, and surrogates across arms. At $n = 2{,}000$, $B = 10$, and 100 replications, the current fixed-$\hat{\ell}$ arm has empirical variance 0.01380 and mean $T = 0.02077$, while refitting $\hat{\ell}$ on every draw has empirical variance 0.01375 and mean $T = 0.01295$. Thus the point estimator is unchanged but T/V falls from 1.505 to 0.942. The difference is almost entirely in $\bar{W}$, which falls from 0.01666 to 0.00876; the between-draw component is unchanged. This isolates reuse of the plug-in outcome nuisance as the source of the stage-5 variance inflation. `R/stage5r_ordinal_refit.R` performs the full correction check at $n = 2{,}000$, $B = 25$, and 200 replications. Bias is 0.016 (Monte Carlo SE 0.007), coverage is 0.960 (Monte Carlo SE 0.014), and $T/V = 1.214$ (Monte Carlo SE 0.122), satisfying the stage's bias, coverage, and variance-calibration criteria. The corrected per-draw refit is therefore the ordinal default.

16 References

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